Network Theory

Previous Year Questions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Network Theory (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

1

Basic Concepts of Networks

Topics

1. Introduction to Electrical Quantities
2. Basic Laws and Classification of Network Elements
3. Resistors, Capacitors and Inductors
4. Energy Sources in Electrical Circuits
5. Series-Parallel Connections
6. Methods of Analyzing Electrical Circuits — Nodal and Mesh Analysis

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Unit Snapshot*

Stream: EC	**Questions:** 31	**MCQ:** 23	**MSQ:** 0	**NAT:** 8
1M: 24	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–6M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 1M

The output voltage V_o (in Volt) for the network given in the Figure is _____. (rounded off to two decimal places)

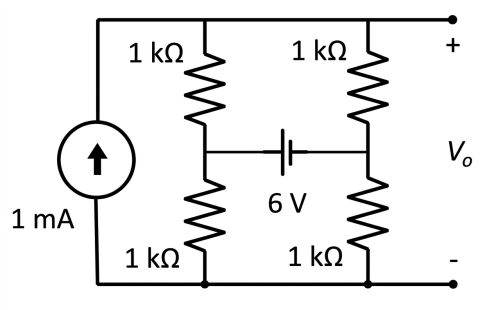


Fig. Q1.26.1

Q1.25.1 MCQ 2025 1M

Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3 Ω resistor are as indicated. The voltage (in Volts) at node X is _____.

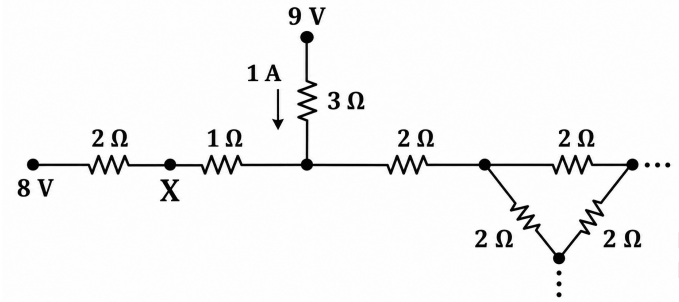


Fig. Q1.25.1

A. $\frac{20}{3}$ **B.** $\frac{32}{3}$ **C.** $\frac{22}{3}$ **D.** $\frac{2}{3}$

Q1.24.1 NAT 2024 1M

In the given circuit, the current I_x (in mA) is _____.

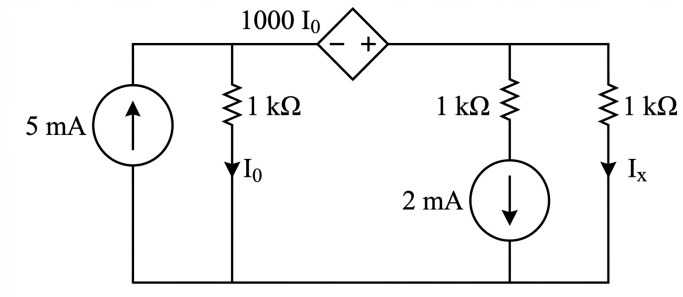


Fig. Q1.24.1

Q1.23.1 NAT 2023 1M

In the circuit shown below, the current i flowing through 200 Ω resistor is _____ mA (rounded off to two decimal places).

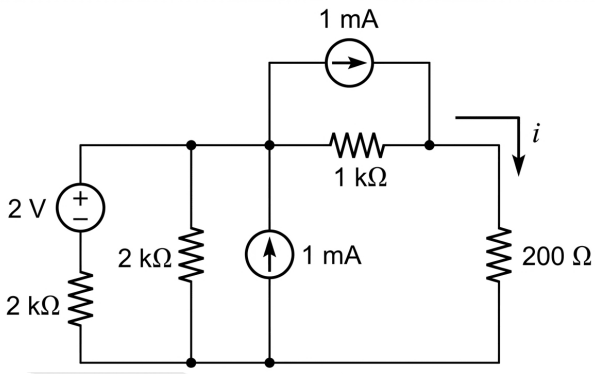


Fig. Q1.23.1

Q1.22.1 MCQ 2022 1M

Consider the circuit shown in the figure. The current I flowing through the 10 Ω resistor is _____.

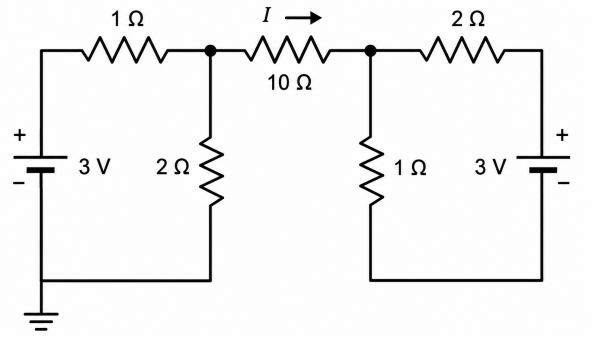


Fig. Q1.22.1

A. 1 A **B.** 0 A **C.** 0.1 A **D.** −0.1 A

Q1.21.1 NAT 2021 1M

Consider the circuit shown in the figure.

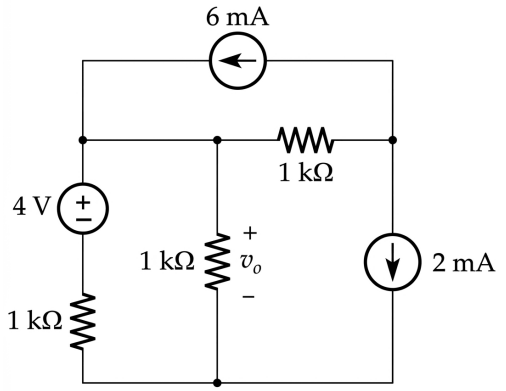


Fig. Q1.21.1

The value of v_0 (rounded off to one decimal place) is _____ V.

Q1.21.2 NAT 2021 1M

Consider the circuit shown in the figure.

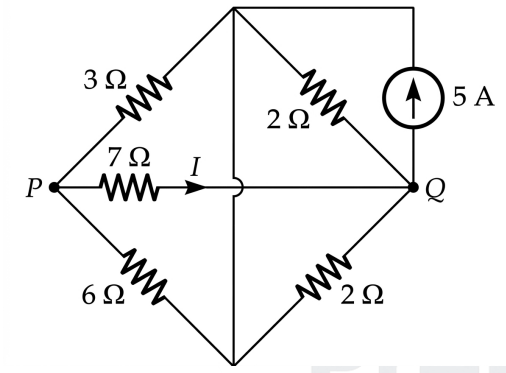


Fig. Q1.21.2

The current 'I' flowing through the 7 Ω resistor between P and Q (Rounded off to 1 decimal place) is _____ A.

Q1.17.1 NAT 2017 1M

A connection is made consisting of resistance A in series with a parallel combination of resistances B and C. Three resistors of value 10 Ω, 5 Ω, 2 Ω are provided. Consider all possible permutations of the given resistors into the positions A, B, C and identify the configurations with maximum possible overall resistance and also the ones with minimum possible overall resistance. The ratio of maximum to minimum values of the resistances (up to two decimal place) is _____.

Q1.16.1 NAT 2016 2M

In the circuit shown in the figure, the magnitude of the current (in amperes) through R_2 is _____.

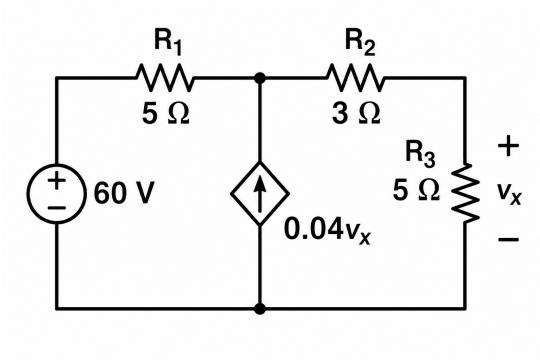


Fig. Q1.16.1

Q1.15.1 NAT 2015 1M

In the circuit shown, the voltage V_x (in Volts) is _____.

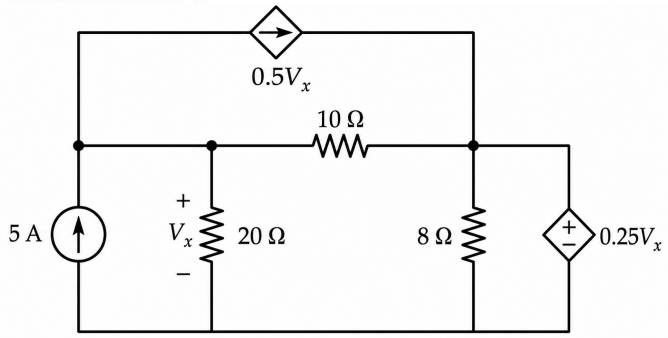


Fig. Q1.15.1

Q1.15.2 MCQ 2015 1M

In the given circuit, the values of V_1 and V_2 respectively are

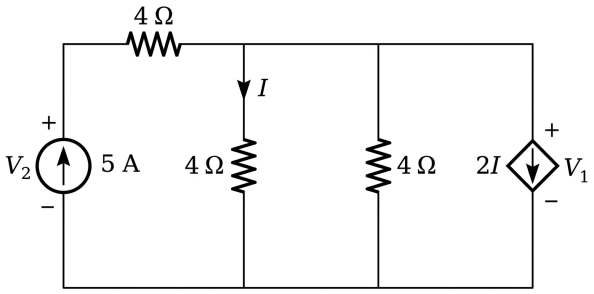


Fig. Q1.15.2

A. 5 V, 25 V
B. 10 V, 30 V
C. 15 V, 35 V
D. 0 V, 20 V

Q1.14.1 MCQ 2014 1M

The circuit shown in the figure represents a

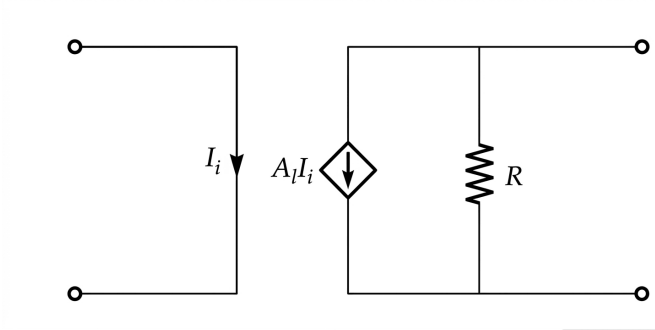


Fig. Q1.14.1

A. voltage controlled voltage source

B. voltage controlled current source

C. current controlled current source

D. current controlled voltage source

Q1.14.2 MCQ 2014 1M

Consider the configuration shown in the figure which is a portion of a larger electrical network.

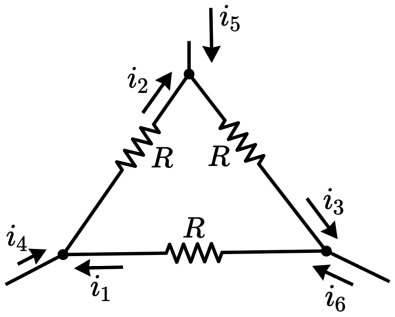


Fig. Q1.14.2

For $R = 1\ \Omega$ and currents $i_1 = 2$ A, $i_4 = -1$ A, $i_5 = -4$ A, which one of the following is TRUE?

A. $i_6 = 5$ A

B. $i_3 = -4$ A

C. Data is sufficient to conclude that the supposed currents are impossible.

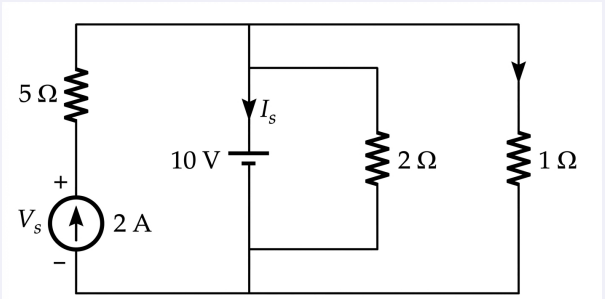
D. Data is insufficient to identify the currents i_2, i_3, and i_6.

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q1.13.1 and Q1.13.2

Consider the following figure:



Q1.13.1 MCQ 2013 2M

The current I_s in amperes in the voltage source, and voltage V_s in volts across the current source respectively, are

A. 13, −20 **B.** 8, −10 **C.** −8, 20 **D.** −13, 20

Q1.13.2 MCQ 2013 2M

The current in the 1 Ω resistor in amperes is

A. 2 **B.** 3.33 **C.** 10 **D.** 12

Q1.13.3 MCQ 2013 2M

Three capacitors C_1, C_2 and C_3 whose values are 10 μF, 5 μF and 2 μF respectively, have breakdown voltages of 10 V, 5 V and 2 V respectively. For the interconnection shown below, the maximum safe voltage in volts that can be applied across the combination, and the corresponding total charge in μC stored in the effective capacitance across the terminals are, respectively

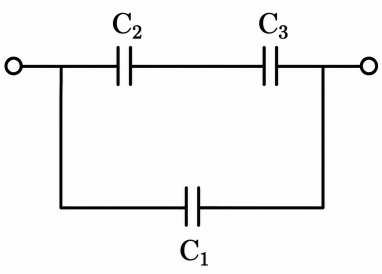


Fig. Q1.13.3

A. 2.8 and 36 **B.** 7 and 119
C. 2.8 and 32 **D.** 7 and 80

Q1.09.1 MCQ 2009 1M

A fully charged mobile phone with a 12 V battery is good for a 10 minute talk-time. Assume that, during the talk-time the battery delivers a constant current of 2 A and its voltage drops linearly from 12 V to 10 V as shown in the figure. How much energy does the battery deliver during this talk-time?

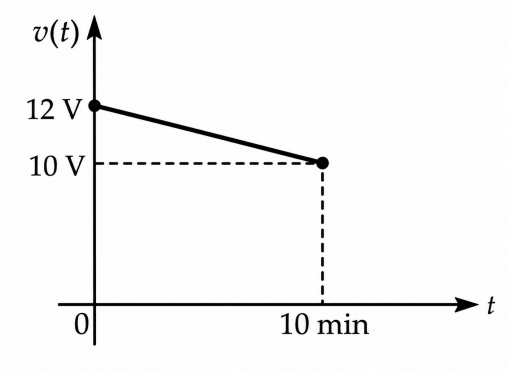


Fig. Q1.09.1

A. 220 J **B.** 12 kJ **C.** 13.2 kJ **D.** 14.4 J

Q1.09.2 MCQ 2009 1M

In the interconnection of ideal sources shown in the below figure, it is known that the 60 V source is absorbing power. Which of the following can be the value of the current source I?

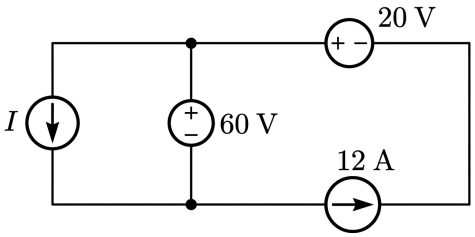


Fig. Q1.09.2

A. 10 A **B.** 13 A **C.** 15 A **D.** 18 A

Q1.05.1 MCQ 2005 2M

If $R_1 = R_2 = R_4 = R$ and $R_3 = 1.1\ R$ in the bridge circuit shown in the figure, then the reading in the ideal voltmeter connected between 'a' and 'b' is

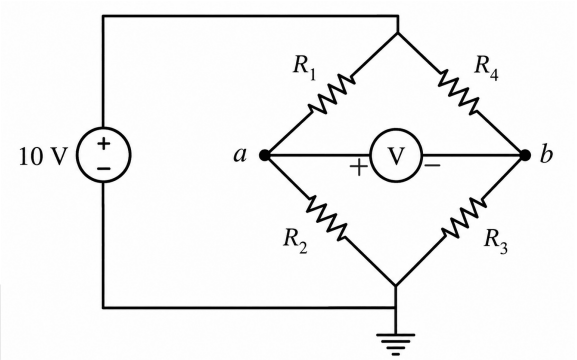


Fig. Q1.05.1

A. 0.238 V **B.** 0.138 V **C.** −0.238 V **D.** 1 V

Q1.02.1 MCQ 2002 1M

The dependent current source shown in the figure,

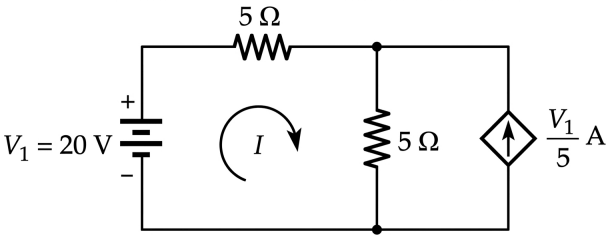


Fig. Q1.02.1

A. delivers 80 W **B.** absorbs 80 W
C. delivers 40 W **D.** absorbs 40 W

Q1.01.1 MCQ 2001 1M

The voltage e_0 in figure is

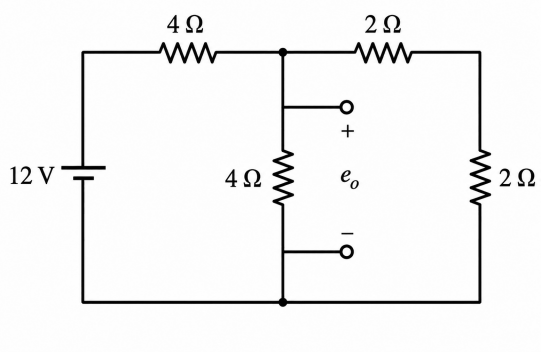


Fig. Q1.01.1

A. 2 V **B.** $\frac{4}{3}$ V **C.** 4 V **D.** 8 V

Q1.00.1 MCQ 2000 1M

In the circuit of the figure, the voltage $v(t)$ is

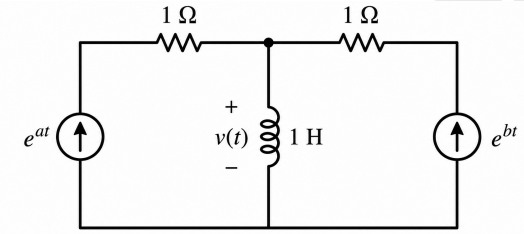


Fig. Q1.00.1

A. $e^{at} - e^{bt}$ **B.** $e^{at} + e^{bt}$
C. $ae^{at} - be^{bt}$ **D.** $ae^{at} + be^{bt}$

Q1.00.2 MCQ 2000 1M

In the circuit of the figure, the value of the voltage source E is

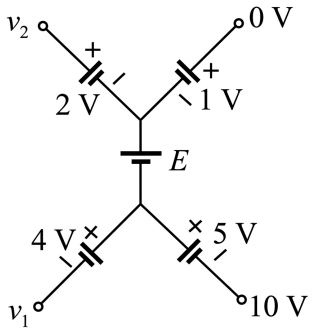


Fig. Q1.00.2

A. −16 V **B.** 4 V **C.** −6 V **D.** 16 V

Q1.98.1 MCQ 1998 1M

The voltage across the terminals 'a' and 'b' in figure is

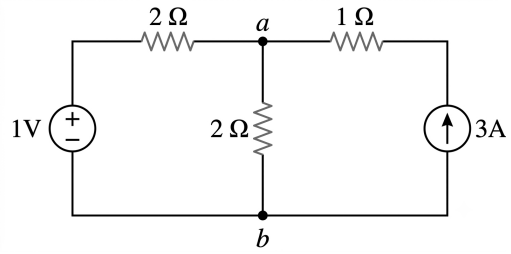


Fig. Q1.98.1

A. 0.5 V **B.** 3.0 V **C.** 3.5 V **D.** 4.0 V

Q1.98.2 MCQ 1998 1M

The nodal method of circuit analysis is based on

A. KVL and Ohm's law

B. KCL and Ohm's law

C. KCL and KVL

D. KCL, KVL and Ohm's law

Q1.97.1 MCQ 1997 1M

The voltage V in figure is

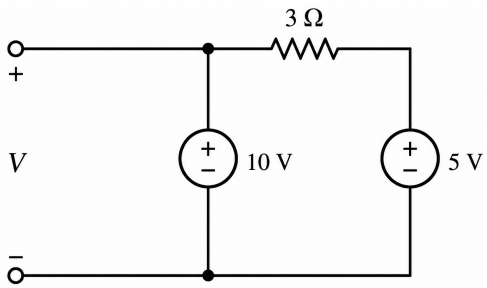


Fig. Q1.97.1

A. 10 V **B.** 15 V **C.** 5 V **D.** None of these

Q1.97.2 MCQ 1997 1M

The voltage V in figure is always equal to

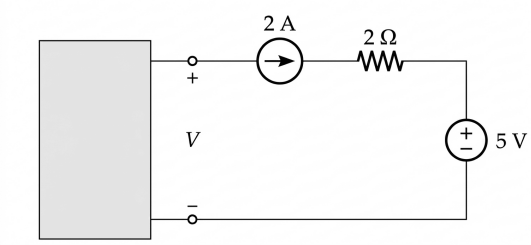


Fig. Q1.97.2

A. 9 V **B.** 5 V **C.** 1 V **D.** None of these

Q1.97.3 MCQ 1997 1M

The voltage V in figure is equal to

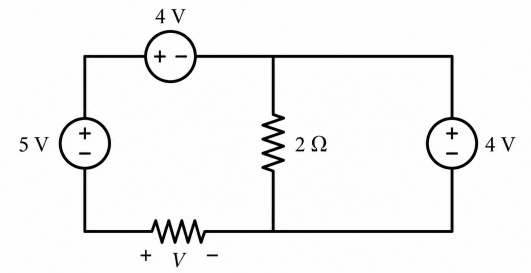


Fig. Q1.97.3

A. 3 V **B.** −3 V **C.** 5 V **D.** None of these

Q1.97.4 MCQ 1997 1M

The current i_4 in the circuit of figure is equal to

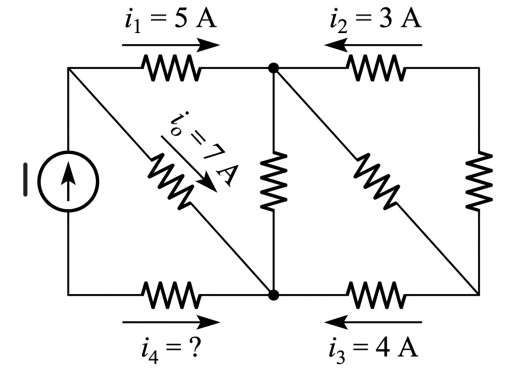


Fig. Q1.97.4

A. 12 A **B.** −12 A **C.** 4 A **D.** None of these

Q1.93.1 MCQ 1993 2M

The two electrical sub-networks N_1 and N_2 are connected through three resistors as shown in figure. The voltage across 5 Ω resistor and 1 Ω resistor are given to be 10 V and 5 V, respectively. Then voltage across 15 Ω resistor is

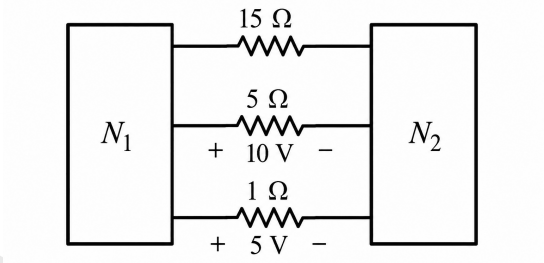


Fig. Q1.93.1

A. −105 V **B.** +105 V **C.** −15 V **D.** +15 V

Q1.93.2 MCQ 1993 2M

A network contains linear resistors and ideal voltage sources. If values of all the resistors are doubled, then the voltage across each resistor is

A. halved **B.** doubled
C. increased by four times **D.** not changed

Answer Key

Q1.26.1 0.95 to 1.05	Q1.25.1 A	Q1.24.1 2
Q1.23.1 1.30 to 1.40	Q1.22.1 B	Q1.21.1 1
Q1.21.2 0.5	Q1.17.1 2.12 to 2.16	Q1.16.1 4.9 to 5.1
Q1.15.1 7.95 to 8.05	Q1.15.2 A	Q1.14.1 C
Q1.14.2 A	Q1.13.1 D	Q1.13.2 C
Q1.13.3 C	Q1.09.1 C	Q1.09.2 A

Q1.05.1 C	Q1.02.1 A	Q1.01.1 C
Q1.00.1 D	Q1.00.2 A	Q1.98.1 C
Q1.98.2 B	Q1.97.1 A	Q1.97.2 D
Q1.97.3 A	Q1.97.4 B	Q1.93.1 A
Q1.93.2 D		

PrepFusion

UNIT

II

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits — Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at $t = 0^+$, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 48	**MCQ:** 34	**MSQ:** 0	**NAT:** 14
1M: 15	**2M:** 33	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–6M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

The relation between the input current (I) and the output voltage (V) of a circuit is governed by;

$$C\frac{dV}{dt} = I(t) - m(t).$$

The circuit is excited by $I(t) = q\delta(t)$, where q is a real-valued constant. The voltage at $t = 0^-$ is V_0. Which of the following is an equivalent representation of the above case?

A. $C\frac{dV}{dt} = -m(t)$, with $V(0^-) = V_0 + \frac{q}{C}$

B. $C\frac{dV}{dt} = -m(t)$, with $V(0^-) = V_0 + \frac{q}{C} + m(0^-)$

C. $C\frac{dV}{dt} = -m(t)$, with $V(0^-) = V_0 - \frac{q}{C} + m(0^-)$

D. $C\frac{dV}{dt} = -m(t)$, with $V(0^-) = \frac{q}{C}$

Q2.24.1 NAT 2024 1M

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____

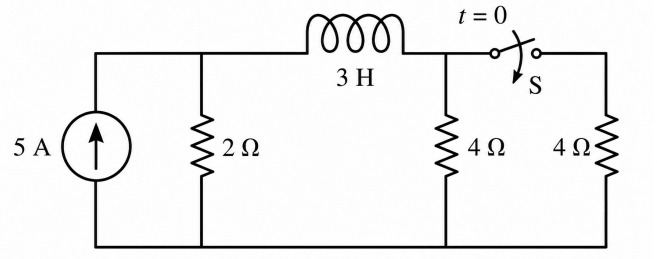


Fig. Q2.24.1

Q2.23.1 NAT 2023 1M

In the circuit shown below, switch S was closed for a long time. If the switch is opened at $t = 0$, the maximum magnitude of the voltage V_R, in volts, is _____ (rounded off to the nearest integer).

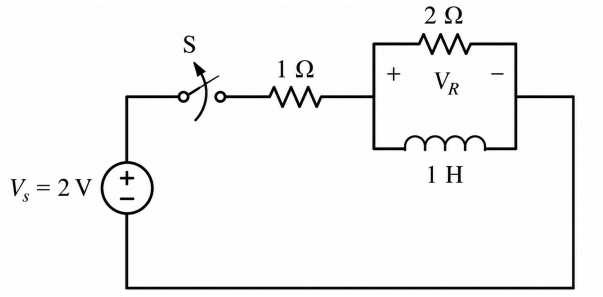


Fig. Q2.23.1

Q2.23.2 MCQ 2023 2M

The switch S_1 was closed and S_2 was open for a long time. At $t = 0$, switch S_1 is opened and S_2 is closed, simultaneously. The value of $i_c(0^+)$, in amperes, is

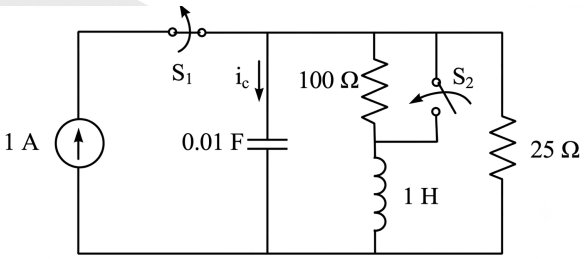


Fig. Q2.23.2

A. 1 **B.** −1 **C.** 0.2 **D.** 0.8

Q2.21.1 NAT 2021 2M

The circuit in the figure contains a current source driving a load having an inductor and a resistor in series, with a shunt capacitor across the load. The ammeter is assumed to have zero resistance. The switch is closed at time, $t = 0$.

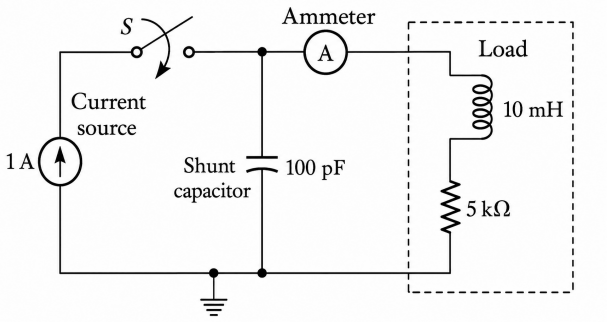


Fig. Q2.21.1

Initially, when the switch is open, the capacitor is discharged and the ammeter reads zero ampere. After the

switch is closed, the ammeter reading keeps fluctuating for some time till it settles to a final steady value. The maximum ammeter reading that one will observe after the switch is closed (Rounded off to two decimal places) is _____ A.

Q2.21.2 NAT 2021 2M

In the circuit shown in the figure, the switch is closed at time $t = 0$, while the capacitor is initially charged to -5 V (i.e., $v_c(0) = -5$ V).

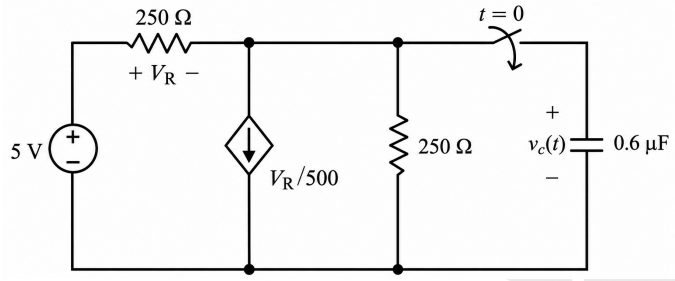


Fig. Q2.21.2

The time after which the voltage across the capacitor becomes zero (rounded off to three decimal places) is _____ ms.

Q2.21.3 MCQ 2021 2M

The switch in the circuit in the figure is in position P for a long time and then moved to position Q at time $t = 0$. The value of $\frac{dv(t)}{dt}$ at $t = 0^+$ is

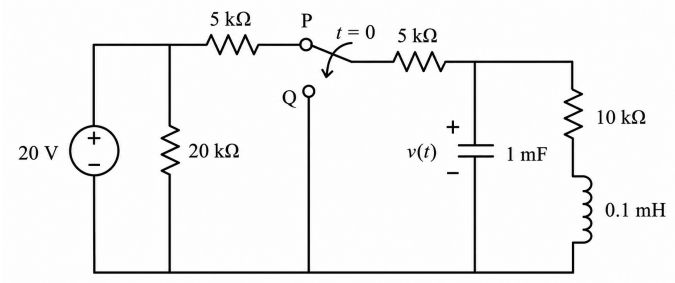


Fig. Q2.21.3

A. 0 V/s **B.** 3 V/s **C.** -3 V/s **D.** -5 V/s

Q2.19.1 NAT 2019 2M

The RC circuit shown below has a variable resistance $R(t)$ given by the following expression:

$$R(t) = R_0\left(1 - \frac{t}{T}\right) \quad \text{for } 0 \le t < T$$

where $R_0 = 1\ \Omega$, and $C = 1$ F. We are also given that $T = 3\ R_0C$ and the source voltage is $V_s = 1$ V. If the current at time $t = 0$ is 1 A. Then the current $I(t)$, in amperes, at time $t = T/2$ is _____. (Rounded off to 2 decimal places).

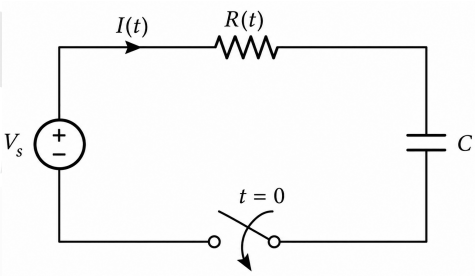


Fig. Q2.19.1

Q2.18.1 NAT 2018 2M

For the circuit given in the figure, the magnitude of the loop current (in amperes, correct to three decimal places) at 0.5 second after closing the switch is _____.

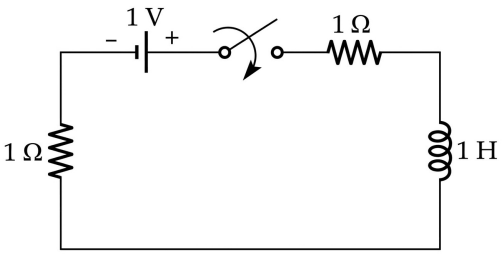


Fig. Q2.18.1

Q2.17.1 NAT 2017 2M

The switch in the circuit, shown in the figure, was open for a long time and is closed at $t = 0$.

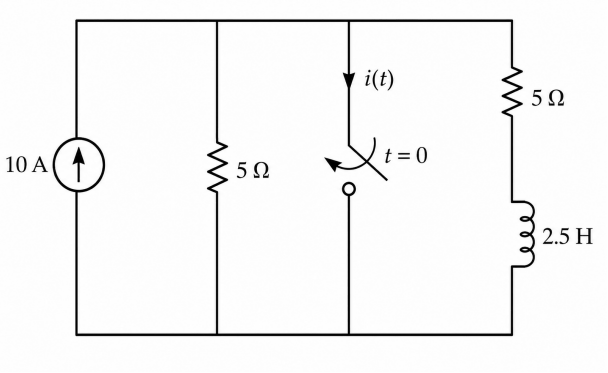


Fig. Q2.17.1

The current $i(t)$ (in Ampere) at $t = 0.5$ seconds is _____.

Q2.17.2 NAT 2017 2M

In the circuit shown, the voltage $V_{IN}(t)$ is described by

$$V_{IN}(t) = \begin{cases} 0, & \text{for } t < 0 \\ 15 \text{ Volts}, & \text{for } t \geq 0 \end{cases}$$

where 't' is in seconds. The time (in seconds) at which the current I in the circuit will reach the value 2 Ampere is _____.

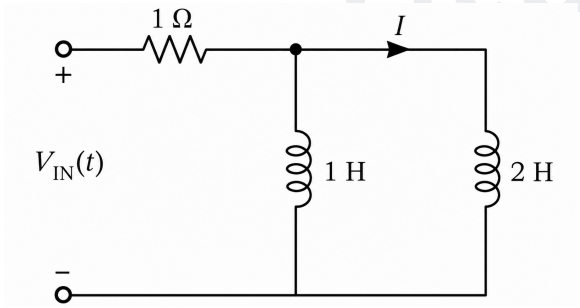


Fig. Q2.17.2

Q2.16.1 NAT 2016 2M

Assume that the circuit in the figure has reached the steady state before time $t = 0$. When the 3 Ω resistor suddenly burns out, resulting in an open circuit. The current $i(t)$ (in ampere) at $t = 0^+$ is _____.

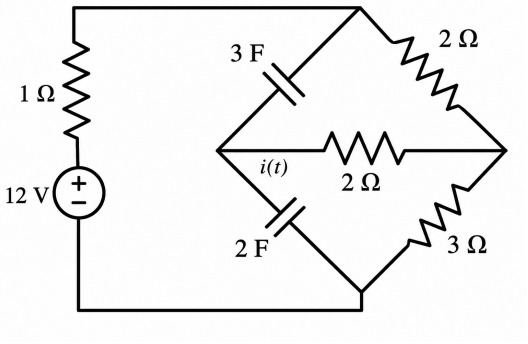


Fig. Q2.16.1

Q2.16.2 NAT 2016 2M

The switch S in the circuit shown has been closed for a long time. It is opened at time $t = 0$ and remains open after that. Assume that the diode has zero reverse current and zero forward voltage drop.

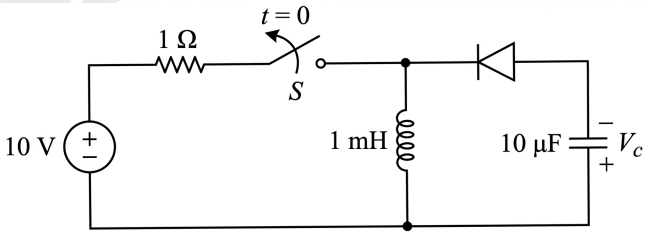


Fig. Q2.16.2

The steady-state magnitude of the capacitor voltage V_C (in Volts), is _____.

Q2.16.3 MCQ 2016 1M

The switch has been in position 1 for a long time and abruptly changes to position 2 at $t = 0$.

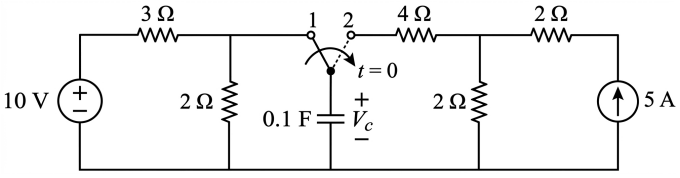


Fig. Q2.16.3

If time 't' is in seconds, the capacitor voltage V_C (in Volts) for $t > 0$ is given by

A. $4\left(1 - \exp\left(-\frac{t}{0.5}\right)\right)$ **B.** $10 - 6\exp\left(-\frac{t}{0.5}\right)$

C. $4\left(1 - \exp\left(-\frac{t}{0.6}\right)\right)$ **D.** $10 - 6\exp\left(-\frac{t}{0.6}\right)$

Q2.15.1 NAT 2015 2M

In the circuit shown, the initial voltages across the capacitors C_1 and C_2 are 1 V and 3 V, respectively. The switch is closed at time $t = 0$. The total energy dissipated (in Joules) in the resistor R until steady-state is reached, is _____.

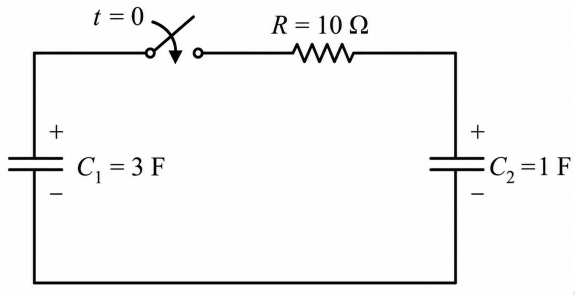


Fig. Q2.15.1

Q2.15.2 NAT 2015 2M

In the circuit shown, switch SW is closed at $t = 0$. Assuming zero initial conditions, the value of $V_c(t)$ (in Volts) at $t = 1$ sec is _____.

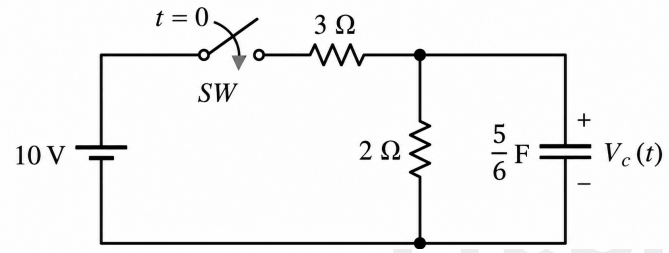


Fig. Q2.15.2

Q2.15.3 MCQ 2015 1M

In the circuit shown, the switch SW is thrown from position A to position B at time $t = 0$. The energy (in μJ) taken from the 3 V source to charge the 0.1 μF capacitor form 0 V to 3 V is

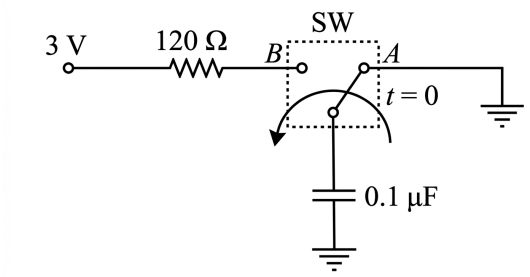


Fig. Q2.15.3

A. 0.3 **B.** 0.45 **C.** 0.9 **D.** 3

Q2.14.1 NAT 2014 2M

In the circuit shown in the figure, the value of $v_o(t)$ (in Volts) for $t \to \infty$ is _____.

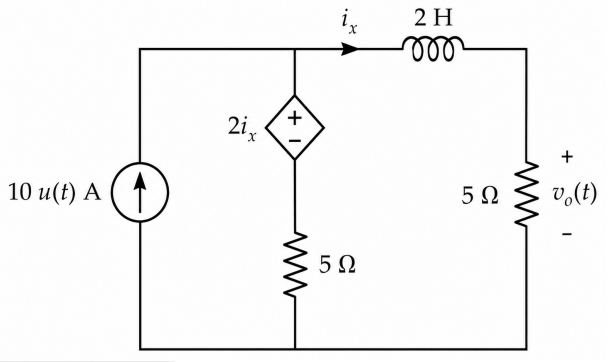


Fig. Q2.14.1

Q2.14.2 MCQ 2014 2M

In the figure shown, the capacitor is initially uncharged. Which one of the following expressions describes the current $I(t)$ (in mA) for $t > 0$?

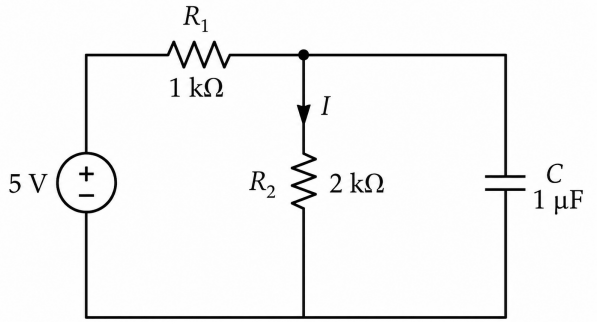


Fig. Q2.14.2

A. $I(t) = \frac{5}{3}(1 - e^{-t/\tau}),\ \tau = \frac{2}{3}$ msec

B. $I(t) = \frac{5}{2}(1 - e^{-t/\tau}),\ \tau = \frac{2}{3}$ msec

C. $I(t) = \frac{5}{3}(1 - e^{-t/\tau}),\ \tau = 3$ msec

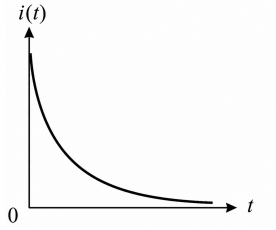
D. $I(t) = \frac{5}{2}(1 - e^{-t/\tau}),\ \tau = 3$ msec

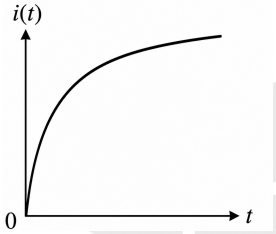
Q2.14.3 MCQ 2014 1M

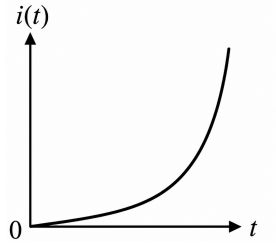
A series RC circuit is connected to a DC voltage source at time $t = 0$. The relation between the source voltage V_s, the resistance R, the capacitance C, and the current $i(t)$ is given below:

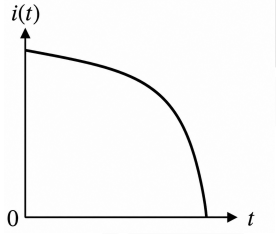
$$V_s = Ri(t) + \frac{1}{C}\int_0^t i(u)\,du$$

Which one of the following represents the current $i(t)$?

A.


B.


C.


D.


Q2.14.4 NAT 2014 1M

In the figure shown, the ideal switch has been open for a long time. If it is closed at $t = 0$, then the magnitude of the current (in mA) through the 4 kΩ resistor at $t = 0^+$ is _____.

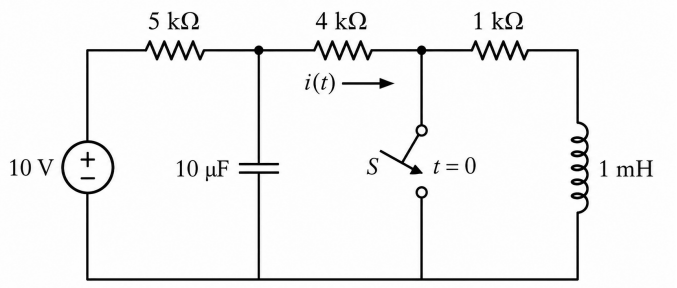


Fig. Q2.14.4

Q2.12.1 MCQ 2012 1M

In the following figure, C_1 and C_2 are ideal capacitors. C_1 had been charged to 12 V before the ideal switch S is closed at $t = 0$. The current $i(t)$ for all 't' is

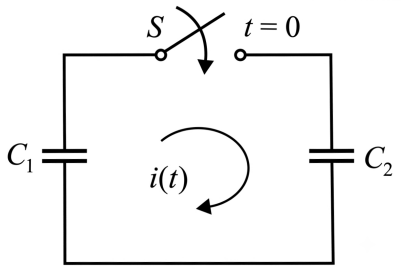


Fig. Q2.12.1

A. zero

B. a step function

C. an exponentially decaying function

D. an impulse function

Q2.11.1 MCQ 2011 2M

In the circuit shown below, the initial charge on the capacitor is 2.5 mC, with the voltage polarity as indicated. The switch is closed at time $t = 0$. The current $i(t)$ at a time 't' after the switch is closed is

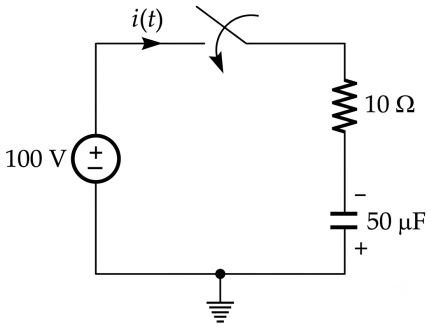


Fig. Q2.11.1

A. $i(t) = 15\exp(-2 \times 10^3 t)$ A

B. $i(t) = 5\exp(-2 \times 10^3 t)$ A

C. $i(t) = 10\exp(-2 \times 10^3 t)$ A

D. $i(t) = -5\exp(-2 \times 10^3 t)$ A

Q2.10.1 MCQ 2010 2M

In the circuit shown, the switch S is open for a long time and is closed at $t = 0$. The current $i(t)$ for $t \geq 0^+$ is

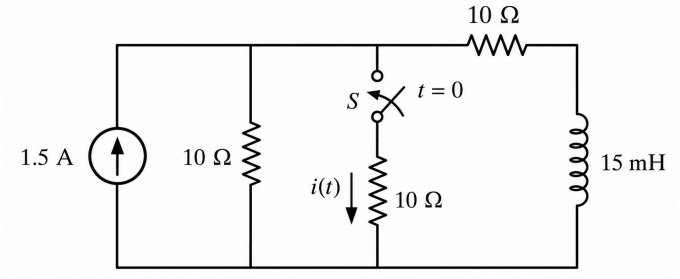


Fig. Q2.10.1

A. $i(t) = 0.5 - 0.125\, e^{-1000t}$ A

B. $i(t) = 1.5 - 0.125\, e^{-1000t}$ A

C. $i(t) = 0.5 - 0.5\, e^{-1000t}$ A

D. $i(t) = 0.375\, e^{-1000t}$ A

Q2.09.1 MCQ 2009 2M

The switch in the circuit shown was on a position a for a long time, and is moved to position 'b' at time $t = 0$. The current $i(t)$ for $t > 0$ is given by

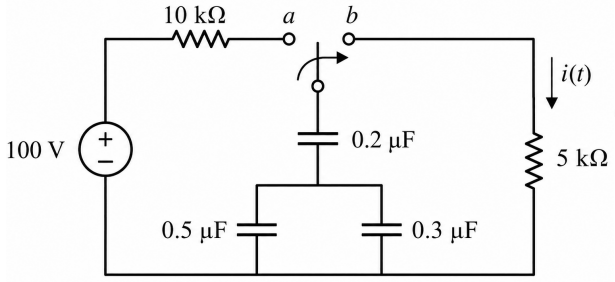


Fig. Q2.09.1

A. $0.2e^{-125t}\, u(t)$ mA

B. $20e^{-1250t}\, u(t)$ mA

C. $0.2e^{-1250t}\, u(t)$ mA

D. $20e^{-1000t}\, u(t)$ mA

Q2.09.2 MCQ 2009 2M

The time domain behaviour of an RL circuit is represented by

$$L\frac{di}{dt} + Ri = V_o\left(1 + Be^{-Rt/L}\sin t\right)u(t)$$

For an initial current of $i(0) = \dfrac{V_o}{R}$, the steady-state value of the current is given by

A. $i(t) \to \dfrac{V_o}{R}$

B. $i(t) \to \dfrac{2V_o}{R}$

C. $i(t) \to \dfrac{V_o}{R}(1 + B)$

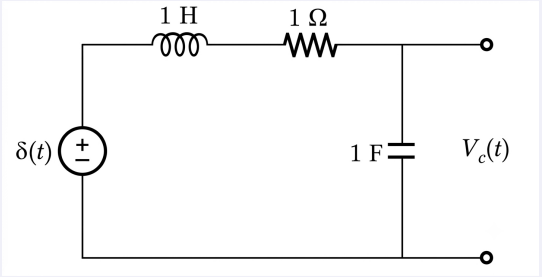
D. $i(t) \to \dfrac{2V_o}{R}(1 + B)$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.08.1 and Q2.08.2

The following series RLC circuit with zero initial conditions is excited by a unit impulse function $\delta(t)$.



Q2.08.1 MCQ 2008 2M

For $t > 0$, the output voltage $V_c(t)$ is

A. $\dfrac{2}{\sqrt{3}}\left(e^{\frac{-1.t}{2}} - e^{-\frac{\sqrt{3}t}{2}}\right)$

B. $\dfrac{2}{\sqrt{3}}\, t\, e^{\frac{-1.t}{2}}$

C. $\dfrac{2}{\sqrt{3}}\, e^{\frac{-1.t}{2}} \cos\left(\dfrac{\sqrt{3}}{2}t\right)$

D. $\dfrac{2}{\sqrt{3}}\, e^{\frac{-1.t}{2}} \sin\left(\dfrac{\sqrt{3}}{2}t\right)$

Q2.08.2 MCQ 2008 2M

For $t > 0$, the voltage across the resistor is

A. $\dfrac{1}{\sqrt{3}}\left(e^{\frac{-\sqrt{3}t}{2}} - e^{\frac{-1.t}{2}}\right)$

B. $e^{\frac{-1}{2}t}\left[\cos\left(\dfrac{\sqrt{3}t}{2}\right) - \dfrac{1}{\sqrt{3}}\sin\left(\dfrac{\sqrt{3}t}{2}\right)\right]$

C. $\dfrac{2}{\sqrt{3}}\, e^{\frac{-1}{2}t} \sin\left(\dfrac{\sqrt{3}t}{2}\right)$

D. $\dfrac{2}{\sqrt{3}}\, e^{\frac{-1}{2}t} \cos\left(\dfrac{\sqrt{3}t}{2}\right)$

Q2.08.3 MCQ 2008 2M

The circuit shown in the figure is used to charge the capacitor C alternately from two current sources as indicated. The switches S_1 and S_2 are mechanically coupled and connected as follows :
For $2nT \le t < (2n+1)T$, $(n = 0, 1, 2, \dots)$ S_1 to P_1 and S_2 to P_2
For $(2n+1)T \le t < (2n+2)T$, $(n = 0, 1, 2, \dots)$ S_1 to Q_1 and S_2 to Q_2.

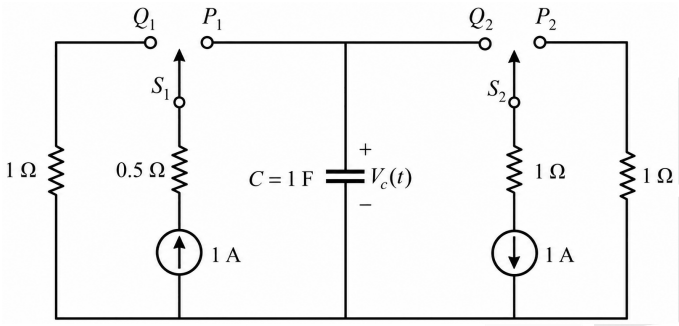


Fig. Q2.08.3

Assume that the capacitor has zero initial charge. Given that $u(t)$ is a unit step function, the voltage $V_C(t)$ across the capacitor is given by

A. $\sum_{n=0}^{\infty}(-1)^n tu(t-nT)$

B. $u(t) + 2\sum_{n=1}^{\infty}(-1)^n u(t-nT)$

C. $tu(t) + 2\sum_{n=1}^{\infty}(-1)^n (t-nT)u(t-nT)$

D. $\sum_{n=0}^{\infty}\left[0.5 - e^{-(t-2nT)} + 0.5e^{-(t-2nT-T)}\right]$

Q2.08.4 MCQ 2008 1M

In the following circuit, the switch S is closed at $t = 0$. The rate of change of current $\frac{di}{dt}(0^+)$ is given by

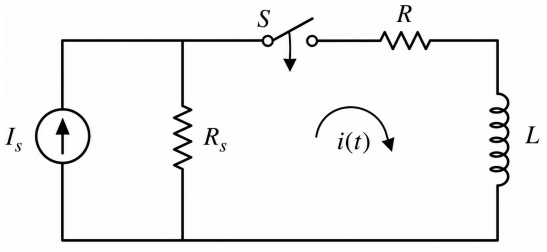


Fig. Q2.08.4

A. 0 **B.** $\frac{R_s I_s}{L}$ **C.** $\frac{(R+R_s)I_s}{L}$ **D.** ∞

Q2.07.1 MCQ 2007 2M

In the circuit shown, V_C is 0 volts at $t = 0$ sec. For $t > 0$, the capacitor current $i_c(t)$, where t is in seconds, is given by

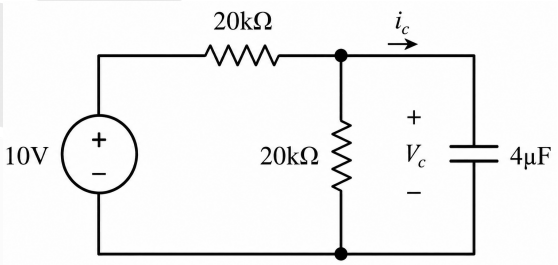


Fig. Q2.07.1

A. $0.50\exp(-25t)$ mA **B.** $0.25\exp(-25t)$ mA
C. $0.50\exp(-12.5t)$ mA **D.** $0.25\exp(-6.25t)$ mA

Q2.06.1 MCQ 2006 1M

In the figure shown below, assume that all the capacitors are initially uncharged. If $v_i(t) = 10\,u(t)$ Volts, $v_o(t)$ is given by

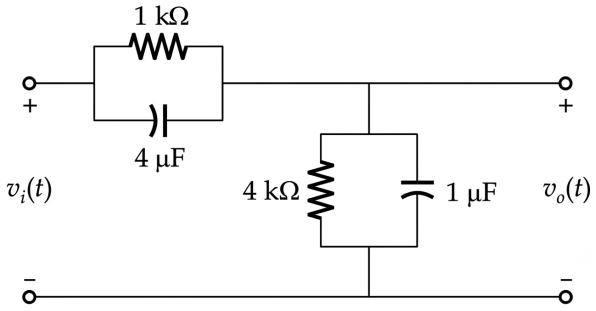


Fig. Q2.06.1

A. $8e^{-t/0.004}$ Volts **B.** $8(1 - e^{-t/0.004})$ Volts
C. $8\,u(t)$ Volts **D.** 8 Volts

Q2.06.2 MCQ 2006 1M

A 2 mH inductor with some initial current can be represented as shown below, where s is the Laplace Transform variable. The value of initial current is

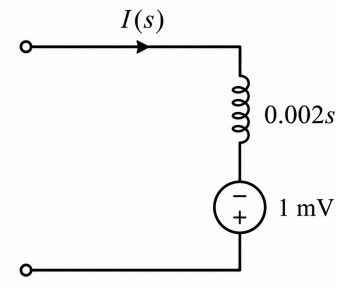


Fig. Q2.06.2

A. 0.5 A **B.** 2.0 A **C.** 1.0 A **D.** 0.0 A

Q2.05.1 MCQ 2005 2M

A square pulse of 3 Volts amplitude is applied to C-R circuit shown in the figure. The capacitor is initially uncharged. The output voltage V_2 at time $t = 2$ sec is

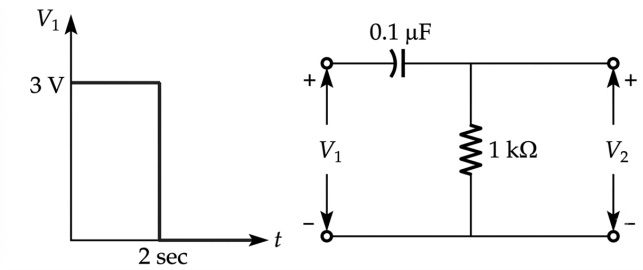


Fig. Q2.05.1

A. 3 V **B.** −3 V **C.** 4 V **D.** −4 V

Q2.04.1 MCQ 2004 2M

The circuit shown in the figure has initial current $i_L(0^-) = 1$ Amp through the inductor and an initial voltage $v_C(0^-) = -1$ V across the capacitor. For input : $v(t) = u(t)$, the Laplace transform of the current $i(t)$ for $t \geq 0$ will be

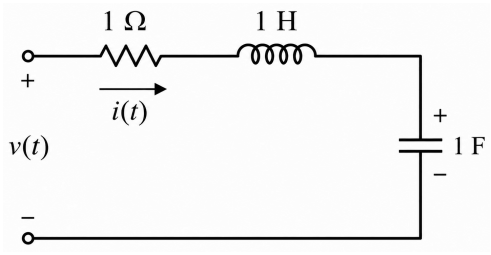


Fig. Q2.04.1

A. $\frac{s}{s^2+s+1}$ **B.** $\frac{s+2}{s^2+s+1}$

C. $\frac{s-2}{s^2+s+1}$ **D.** $\frac{s+1}{s^2+s+1}$

Q2.04.2 MCQ 2004 1M

For the R-L circuit shown in figure, the input voltage $v_i(t) = u(t)$. The current $i(t)$ is

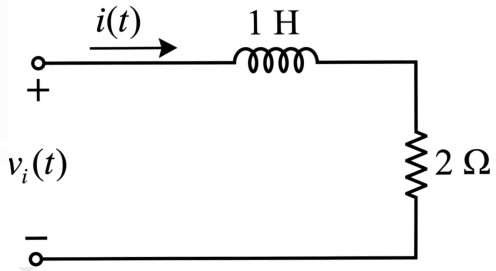
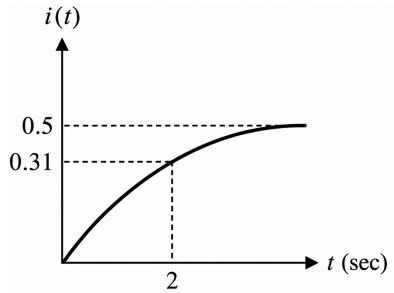
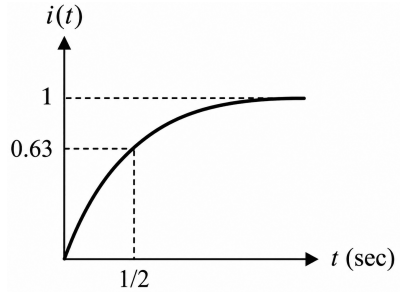


Fig. Q2.04.2

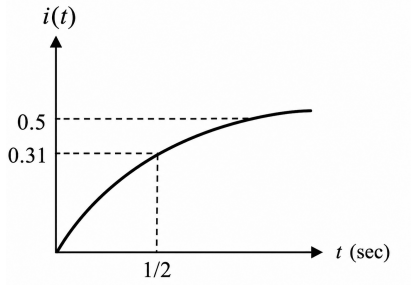
A.



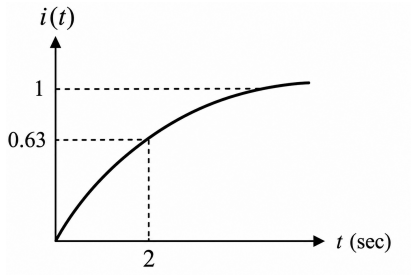
B.



C.



D.

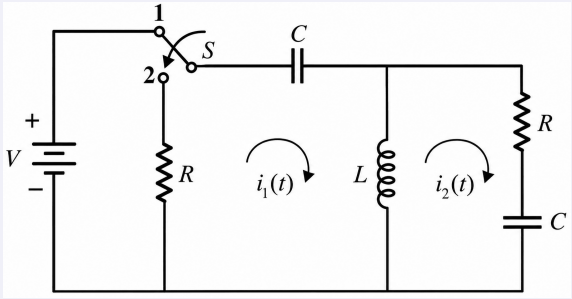


Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions Q2.03.1 and Q2.03.2

Assume that the switch S is in position 1 for a long time and thrown to position 2 at t = 0.



Q2.03.1 MCQ 2003 2M

At $t = 0^+$, the current i_1 is

A. $\frac{-V}{2R}$ **B.** $\frac{-V}{R}$ **C.** $\frac{-V}{4R}$ **D.** zero

Q2.03.2 MCQ 2003 2M

$I_1(s)$ and $I_2(s)$ are the Laplace transforms of $i_1(t)$ and $i_2(t)$ respectively. The equations for the loop currents $I_1(s)$ and $I_2(s)$ for the circuit shown in the figure, after the switch is brought from position 1 to position 2 at $t = 0$, are

A. $\begin{bmatrix} R + Ls + \frac{1}{Cs} & -Ls \\ -Ls & R + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V/s \\ 0 \end{bmatrix}$

B. $\begin{bmatrix} R + Ls + \frac{1}{Cs} & -Ls \\ -Ls & R + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} -V/s \\ 0 \end{bmatrix}$

C. $\begin{bmatrix} R + Ls + \frac{1}{Cs} & -Ls \\ -Ls & R + Ls + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} -V/s \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} R + Ls + \frac{1}{Cs} & -Ls \\ -Ls & R + Ls + \frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V/s \\ 0 \end{bmatrix}$

Q2.02.1 MCQ 2002 1M

In the figure, the switch was closed for a long time before opening at $t = 0$. The voltage V_x at $t = 0^+$ is,

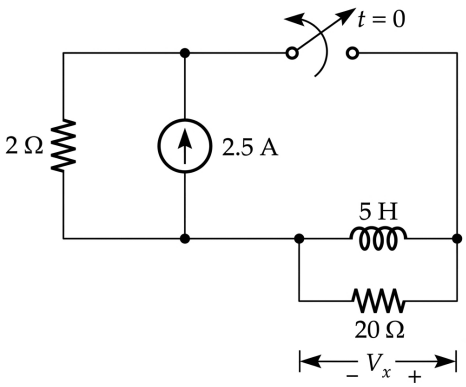


Fig. Q2.02.1

A. 25 V **B.** 50 V **C.** −50 V **D.** 0 V

Q2.97.1 MCQ 1997 2M

In the circuit of figure the energy absorbed by the 4 Ω resistor in time interval $(0, \infty)$ is [Given : $V_C(0) = 6V$]

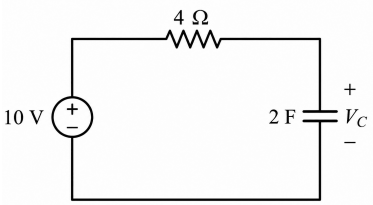


Fig. Q2.97.1

A. 36 J **B.** 16 J **C.** 256 J **D.** None of these

Q2.96.1 MCQ 1996 2M

The voltage V_{C_1}, V_{C_2} and V_{C_3} across the capacitors in the circuit in figure, under steady-state, are respectively

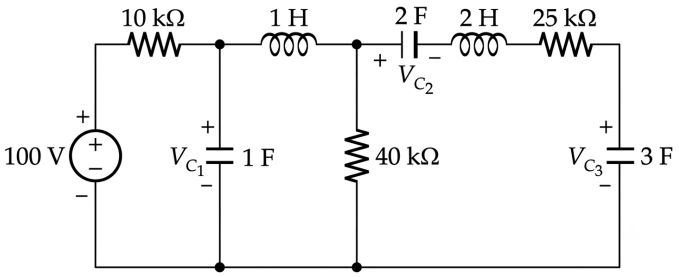


Fig. Q2.96.1

A. 80 V, 32 V, 48 V
B. 80 V, 48 V, 32 V
C. 20 V, 8 V, 12 V
D. 20 V, 12 V, 8 V

Q2.95.1 MCQ 1995 1M

Consider a DC voltage source connected to a series *R*-*C* circuit. When the steady-state reaches, the ratio of the energy stored in the capacitor to the total energy supplied by the voltage source, is equal to

A. 0.362 **B.** 0.500 **C.** 0.632 **D.** 1.000

Q2.94.1 MCQ 1994 1M

A ramp voltage, $v(t) = 100t$ Volts, is applied to an *RC* differentiating circuit with $R = 5$ kΩ and $C = 4$ μF. The maximum output voltage is

A. 0.2 volt
B. 2.0 volts
C. 10.0 volts
D. 50.0 volts

Q2.92.1 MCQ 1992 2M

For the circuit shown below different time constants are given

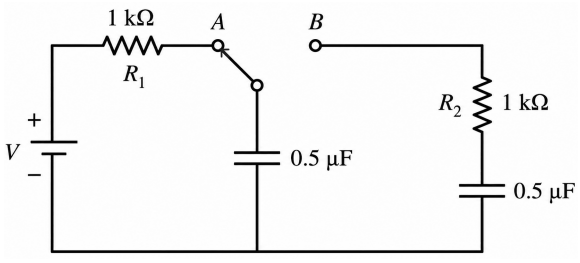


Fig. Q2.92.1

(1) 0.5×10^{-3} sec
(2) 2×10^{-3} sec
(3) 0.25×10^{-3} sec
(4) 10^{-3} sec

What are the charging and discharging times respectively

A. 1, 2 **B.** 2, 3 **C.** 1, 3 **D.** 2, 4

Q2.92.2 MCQ 1992 2M

For the compensated attenuator of figure, the impulse response under the condition $R_1C_1 = R_2C_2$ is

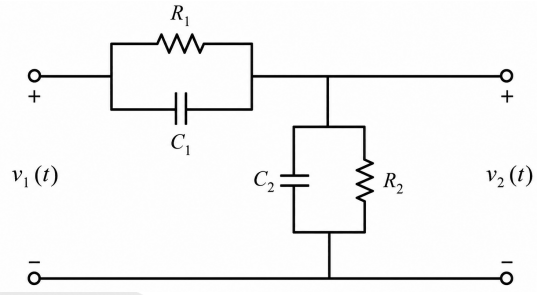


Fig. Q2.92.2

A. $\frac{R_2}{R_1 + R_2}[1 - e^{-\frac{1}{R_1C_1}t}]u(t)$
B. $\frac{R_2}{R_1 + R_2}\delta(t)$
C. $\frac{R_2}{R_1 + R_2}u(t)$
D. $\frac{R_2}{R_1 + R_2}e^{-\frac{1}{R_1C_1}t}u(t)$

Q2.89.1 MCQ 1989 2M

If the Laplace transform of the voltage across a capacitor of value of $\frac{1}{2}$ F is,

$$V_C(s) = \frac{s+1}{s^3 + s^2 + s + 1}$$

The value of current through the capacitor at $t = 0^+$ is

A. 0 A **B.** 2 A **C.** (1/2) A **D.** 1 A

Q2.89.2 MCQ 1989 2M

A 10 Ω resistor, a 1 H inductor and 1 μF capacitor are connected in parallel. The combination is driven by a unit step current. Under the steady state condition, the source current flows through

A. the resistor
B. the inductor
C. the capacitor
D. all the three elements

Q2.87.1 MCQ 1987 2M

A square waveform as shown in figure is applied across 1 mH ideal inductor. The current through inductor is a

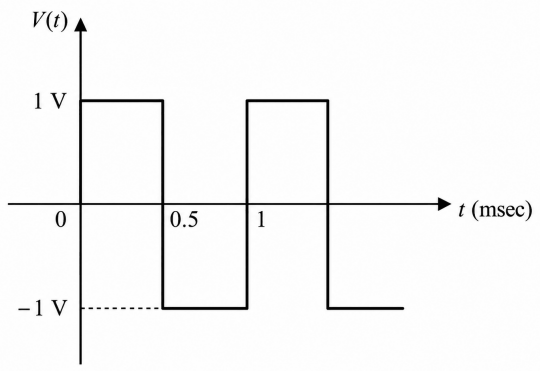


Fig. Q2.87.1

A. triangular wave with peak amplitude of 0.5 Amp.

B. triangular wave with peak amplitude of 1.0 Amp.

C. triangular wave with peak amplitude of 2.0 Amp.

D. sine wave with peak amplitude of 0.5 Amp.

Q2.14.2 A	Q2.14.3 A	Q2.14.4 1.20 to 1.30
Q2.12.1 D	Q2.11.1 A	Q2.10.1 A
Q2.09.1 B	Q2.09.2 A	Q2.08.1 D
Q2.08.2 B	Q2.08.3 C	Q2.08.4 B
Q2.07.1 A	Q2.06.1 C	Q2.06.2 A
Q2.05.1 B	Q2.04.1 B	Q2.04.2 C
Q2.03.1 A	Q2.03.2 C	Q2.02.1 C
Q2.97.1 B	Q2.96.1 B	Q2.95.1 B
Q2.94.1 B	Q2.92.1 C	Q2.92.2 B
Q2.89.1 C	Q2.89.2 B	Q2.87.1 A

Answer Key

Q2.26.1 A	Q2.24.1 0.75	Q2.23.1 4
Q2.23.2 B	Q2.21.1 1.40 to 1.50	Q2.21.2 0.132 to 0.146
Q2.21.3 C	Q2.19.1 0.23 to 0.27	Q2.18.1 0.284 to 0.348
Q2.17.1 8.00 to 8.30	Q2.17.2 0.30 to 0.40	Q2.16.1 -1.02 to -0.98 or 0.98 to 1.02 or 1.96 to 2.04 or -2.04 to -1.96
Q2.16.2 99 to 101	Q2.16.3 D	Q2.15.1 1.4 to 1.6
Q2.15.2 2.480 to 2.580	Q2.15.3 C	Q2.14.1 31.24 to 31.26

UNIT

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and Conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Unit Snapshot*

Stream: EC	**Questions:** 39	**MCQ:** 24	**MSQ:** 1	**NAT:** 14
1M: 17	**2M:** 22	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–3M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Q3.24.1 NAT 2024 2M

In the network shown below, maximum power is to be transferred to the load R_L

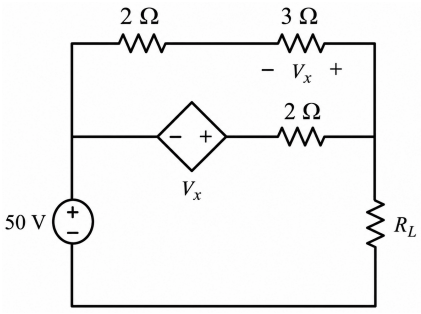


Fig. Q3.24.1

The value of R_L (in Ω) is _____.

Q3.22.1 MCQ 2022 1M

The current I in the circuit shown is _____.

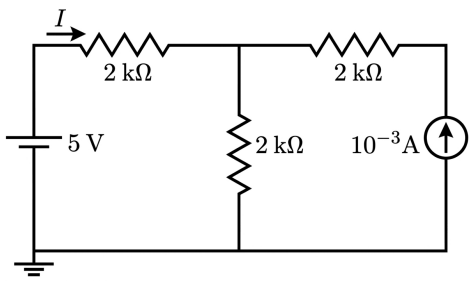


Fig. Q3.22.1

A. 1.25×10^{-3}A **B.** 0.75×10^{-3}A
C. -0.5×10^{-3}A **D.** 1.16×10^{-3}A

Q3.20.1 MCQ 2020 1M

In the circuit shown below, the Thevenin voltage V_{Th} is

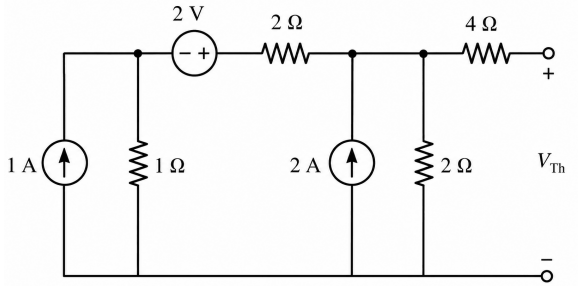


Fig. Q3.20.1

A. 2.8 V **B.** 3.6 V **C.** 2.4 V **D.** 4.5 V

Q3.19.1 MCQ 2019 1M

Consider the two-port resistive network shown in the figure. When an excitation of 5 V is applied across Port 1, and Port 2 is shorted, the current through the short circuit at Port 2 is measured to be 1 A (see (a) in the figure).
Now, if an excitation of 5 V is applied across Port 2, and Port 1 is shorted (see (b) in the figure), what is the current through the short circuit at Port 1?

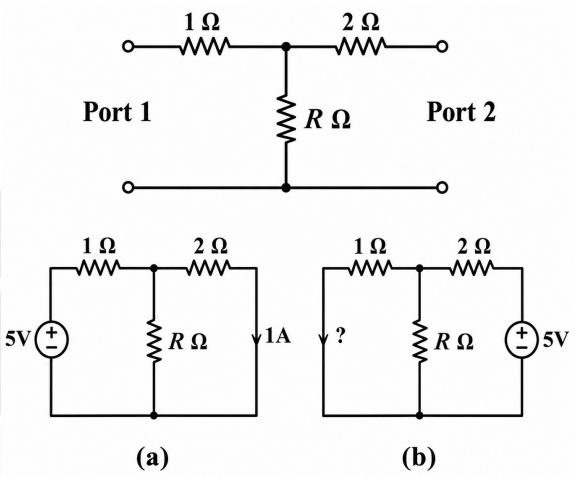


Fig. Q3.19.1

A. 0.5 A **B.** 1.0 A **C.** 2.0 A **D.** 2.5 A

Q3.18.1 NAT 2018 2M

Consider the network shown below with $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$ and $R_3 = 3\ \Omega$. The network is connected to a constant voltage source of 11V.

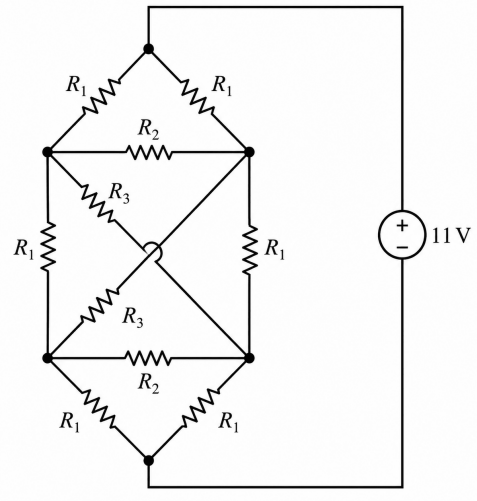


Fig. Q3.18.1

The magnitude of the current (in amperes, accurate to two decimal places) through the source is _____.

Q3.17.1 NAT 2017 2M

Consider the circuit shown in the figure.

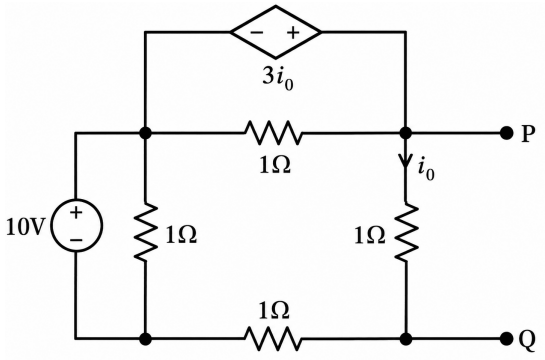


Fig. Q3.17.1

The Thevenin equivalent resistance (in Ω) across P-Q is _____.

Q3.16.1 MCQ 2016 1M

In the circuit shown below, V_s is a constant voltage source and I_L is a constant load current.

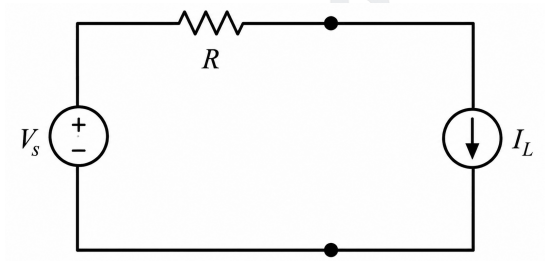


Fig. Q3.16.1

The value of I_L that maximizes the power absorbed by the constant current load is

A. $\frac{V_s}{4R}$ **B.** $\frac{V_s}{2R}$ **C.** $\frac{V_s}{R}$ **D.** ∞

Q3.16.2 NAT 2016 2M

In the circuit shown in the figure, the maximum power (in Watt) delivered to the resistor R is _____.

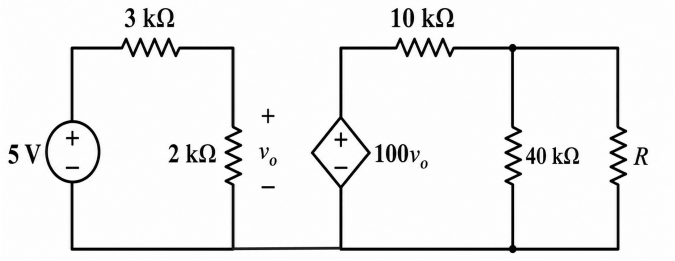


Fig. Q3.16.2

Q3.16.3 NAT 2016 2M

In the figure shown, the current i (in ampere) is _____.

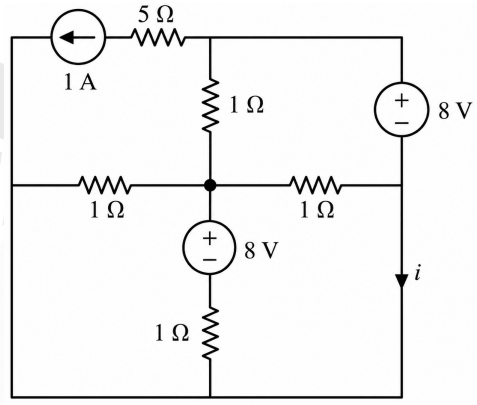


Fig. Q3.16.3

Q3.16.4 MCQ 2016 2M

In the given circuit, each resistor has a value equal to 1 Ω.

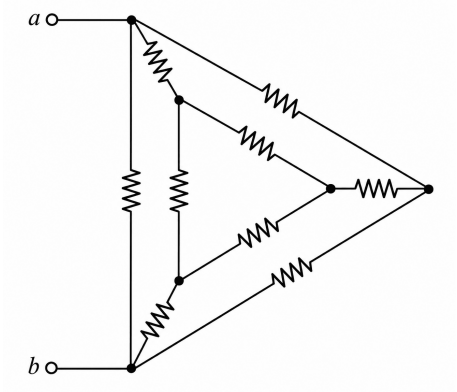


Fig. Q3.16.4

What is the equivalent resistance across the terminals 'a' and 'b'?

A. $\frac{1}{6}$ Ω **B.** $\frac{1}{3}$ Ω **C.** $\frac{9}{20}$ Ω **D.** $\frac{8}{15}$ Ω

Q3.15.1 NAT 2015 1M

For the circuit shown in the figure, the Thevenin's equivalent voltage (in Volts) across terminals $a - b$ is _____.

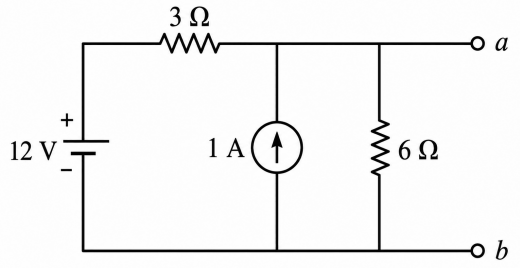


Fig. Q3.15.1

Q3.15.2 NAT 2015 2M

In the circuit shown, the Norton equivalent resistance (in Ω) across terminals $a - b$ is _____.

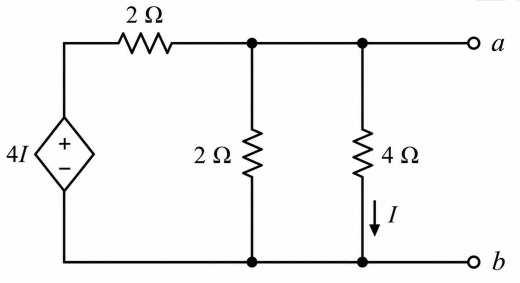


Fig. Q3.15.2

Q3.15.3 NAT 2015 1M

In the network shown in the figure, all resistors are identical with $R = 300\ \Omega$. The resistance R_{ab} (in Ω) of the network is _____.

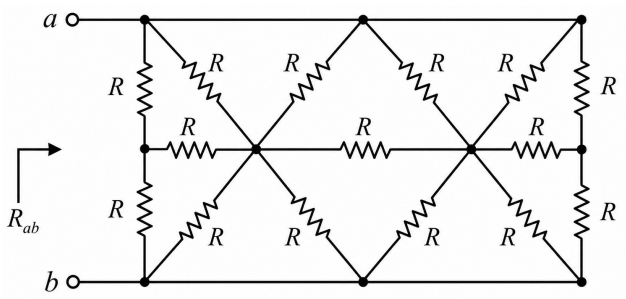


Fig. Q3.15.3

Q3.14.1 MCQ 2014 1M

Norton's theorem states that a complex network connected to a load can be replaced with an equivalent impedance

A. in series with a current source.

B. in parallel with a voltage source.

C. in series with a voltage source.

D. in parallel with a current source.

Q3.14.2 MCQ 2014 1M

For maximum power transfer between two cascaded sections of an electrical network, the relationship between the output impedance Z_1 of the first section to the input impedance Z_2 of the second section is

A. $Z_2 = Z_1$ **B.** $Z_2 = -Z_1$

C. $Z_2 = Z_1^*$ **D.** $Z_2 = -Z_1^*$

Q3.14.3 NAT 2014 2M

The equivalent resistance in the infinite ladder network shown in the figure, is R_e.

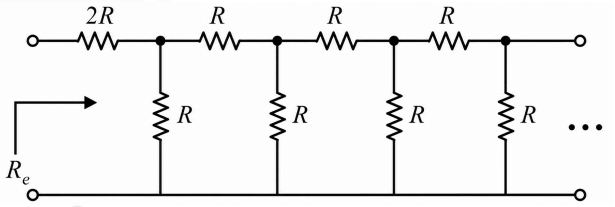


Fig. Q3.14.3

The value of $\frac{R_e}{R}$ is _____.

Q3.14.4 NAT 2014 1M

The magnitude of current (in mA) through the resistor R_2 in the figure shown is _____.

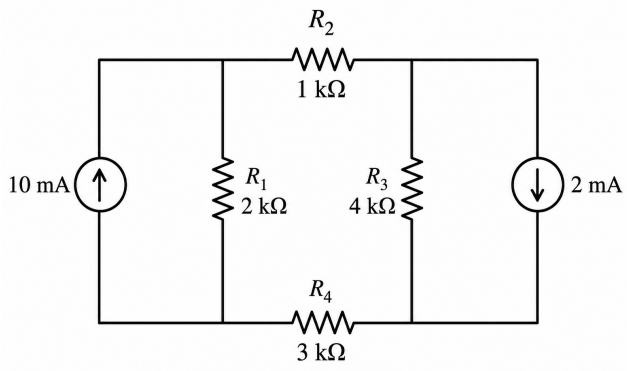


Fig. Q3.14.4

Q3.14.5 NAT 2014 1M

In the figure shown, the value of current I (in Amperes) is _____.

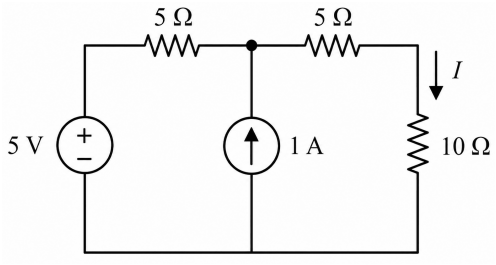


Fig. Q3.14.5

Q3.14.6 NAT 2014 2M

A Y-network has resistance of 10 Ω each in two of its arms, while the third arm has a resistance of 11 Ω. In the equivalent Δ-network, the lowest value (in Ω) among the three resistances is _____.

Q3.14.7 NAT 2014 2M

For the Y-network shown in the figure, the value of R_1 (in Ω) in the equivalent Δ-network is _____.

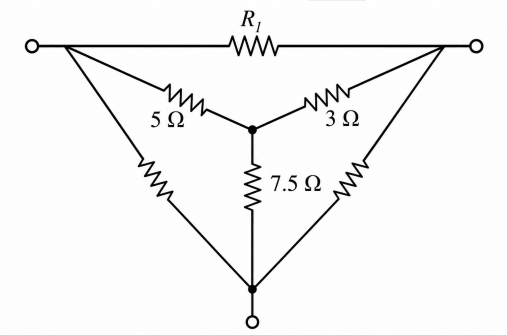


Fig. Q3.14.7

Q3.13.1 MCQ 2013 1M

Consider a delta connection of resistors and its equivalent star connection is shown below. If all elements of the delta connection are scaled by a factor k, $k > 0$, the elements of the corresponding star equivalent will be scaled by a factor of

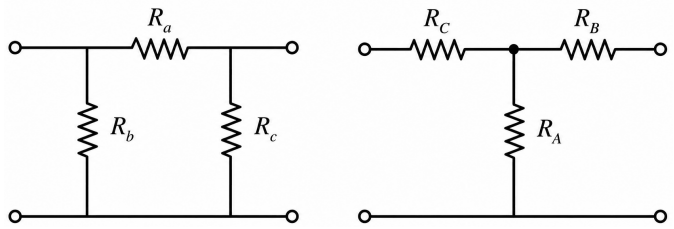


Fig. Q3.13.1

A. k^2 **B.** k **C.** $1/k$ **D.** $\sqrt{k}$

Q3.12.1 MCQ 2012 2M

If $V_A - V_B = 6$ V, then $V_C - V_D$ is

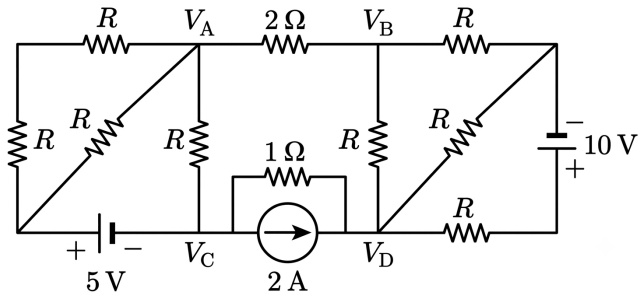


Fig. Q3.12.1

A. −5 V **B.** 2 V **C.** 3 V **D.** 6 V

Q3.12.2 MCQ 2012 1M

The impedance looking into nodes 1 and 2 in the given circuit is

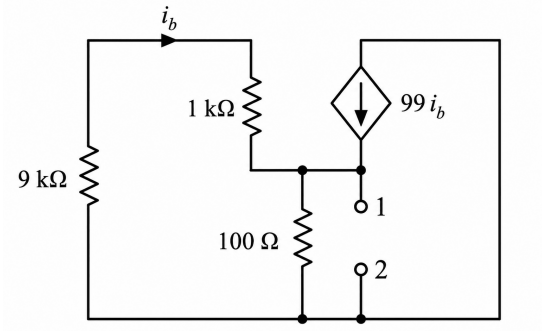


Fig. Q3.12.2

A. 50 Ω **B.** 100 Ω **C.** 5 kΩ **D.** 10.1 kΩ

Q3.11.1 MCQ 2011 1M

In the circuit shown below, the value of R_L such that the power transferred to R_L is maximum

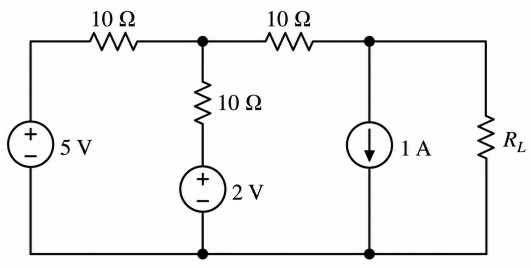


Fig. Q3.11.1

A. 5 Ω **B.** 10 Ω **C.** 15 Ω **D.** 20 Ω

Q3.10.1 MCQ 2010 2M

In the circuit shown, the power supplied by the voltage source is

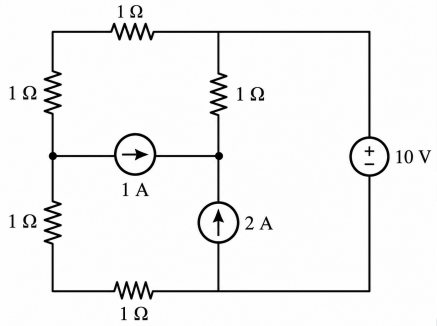


Fig. Q3.10.1

A. 0 W **B.** 5 W **C.** 10 W **D.** 100 W

Q3.09.1 MCQ 2009 2M

In the circuit shown, what value of R_L maximizes the power delivered to R_L?

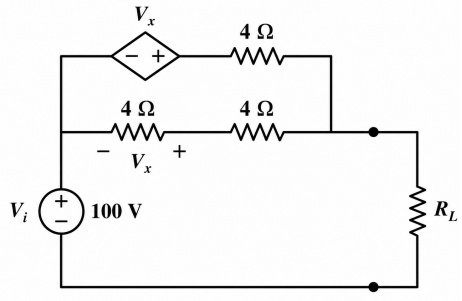


Fig. Q3.09.1

A. 2.4 Ω **B.** 3.66 Ω **C.** 4 Ω **D.** 6 Ω

Q3.08.1 MCQ 2008 2M

The Thevenin equivalent impedance Z_{TH} between the nodes P and Q in the following circuit is

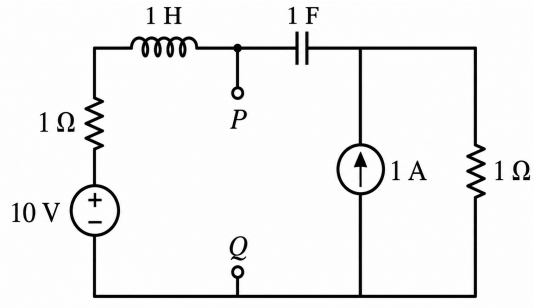


Fig. Q3.08.1

A. 1 **B.** $1 + s + \frac{1}{s}$

C. $2 + s + \frac{1}{s}$ **D.** $\frac{s^2 + s + 1}{s^2 + 2s + 1}$

Q3.07.1 MCQ 2007 2M

For the circuit shown in the figure, the Thevenin voltage and resistance looking into X-Y are

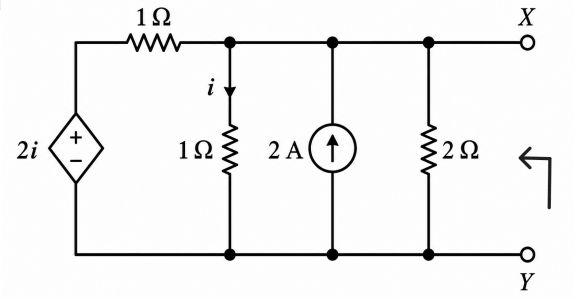


Fig. Q3.07.1

A. 4/3 V, 2 Ω **B.** 4 V, 2/3 Ω
C. 4/3 V, 2/3 Ω **D.** 4 V, 2 Ω

Q3.05.1 MCQ 2005 2M

For the circuit shown in the figure, Thevenin's voltage and Thevenin's equivalent resistance at terminals a-b is

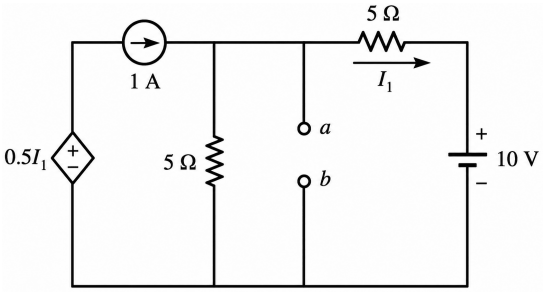


Fig. Q3.05.1

A. 5 V and 2 Ω **B.** 7.5 V and 2.5 Ω
C. 4 V and 2 Ω **D.** 3 V and 2.5 Ω

Q3.05.2 MCQ 2005 1M

The maximum power that can be transferred to the load resistor R_L from the voltage source in the figure is

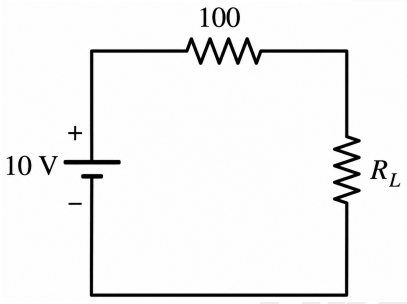


Fig. Q3.05.2

A. 1 W **B.** 10 W **C.** 0.25 W **D.** 0.5 W

Q3.03.1 MCQ 2003 2M

Twelve 1 Ω resistances are used as edges to form a cube. The resistance between two diagonally opposite corners of the cube is

A. $\frac{5}{6}$ Ω **B.** 1 Ω **C.** $\frac{6}{5}$ Ω **D.** $\frac{3}{2}$ Ω

Q3.02.1 MCQ 2002 2M

In the network of the figure, the maximum power is delivered to R_L if its value is

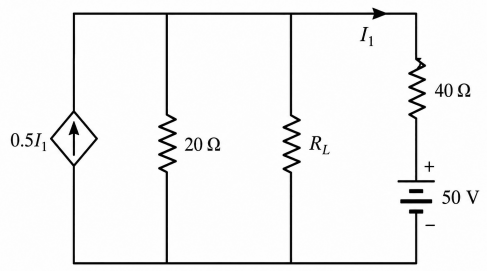


Fig. Q3.02.1

A. 16 Ω **B.** $\frac{40}{3}$ Ω **C.** 60 Ω **D.** 20 Ω

Q3.01.1 MCQ 2001 2M

The voltage e_o in the figure is

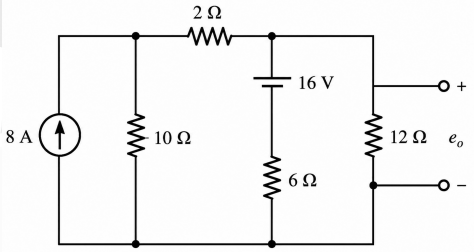


Fig. Q3.01.1

A. 48 V **B.** 24 V **C.** 36 V **D.** 28 V

Q3.00.1 MCQ 2000 2M

Use the data of the Fig. (a). The current 'i' in the circuit of the Fig. (b).

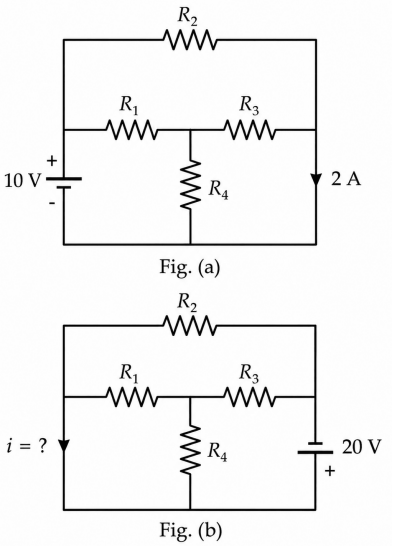


Fig. Q3.00.1

A. −2 A **B.** 2 A **C.** −4 A **D.** +4 A

Q3.99.1 MCQ 1999 2M

The value of R (in ohms) required for maximum power transfer in the network shown in figure is

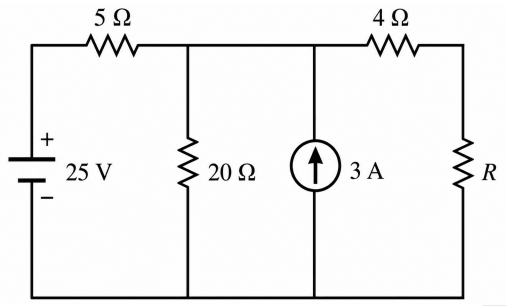


Fig. Q3.99.1

A. 2 **B.** 4 **C.** 8 **D.** 16

Q3.98.1 MCQ 1998 1M

Superposition theorem is not applicable to networks containing

A. non-linear elements

B. dependent voltage sources

C. dependent current sources

D. transformers

Q3.95.1 MSQ 1995 1M

A dc circuit shown in figure has a voltage source V, a current source I and several resistors. A particular resistor R dissipates a power of 4 Watts when V alone is active. The same resistor R dissipates a power of 9 Watts when I alone is active. The power dissipated by R when both sources are active will be

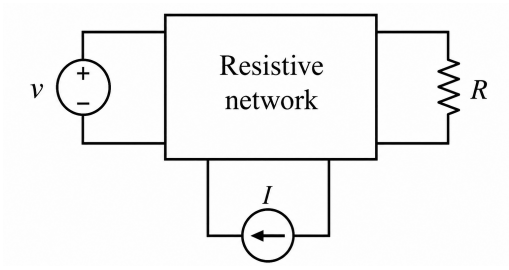


Fig. Q3.95.1

A. 1 W **B.** 5 W **C.** 13 W **D.** 25 W

Q3.95.2 MCQ 1995 1M

The value of the resistance, R connected across the terminals, A and B (ref. figure), which will absorb the maximum power, is

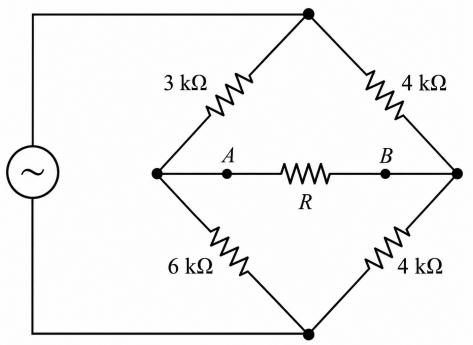


Fig. Q3.95.2

A. 4.00 kΩ **B.** 4.11 kΩ **C.** 8.00 kΩ **D.** 9.00 kΩ

Q3.93.1 NAT 1993 2M

In the circuit of figure, when switch S_1 is closed, the ideal ammeter M_1 reads 5 A. What will the ideal voltmeter M_2 read when S_1 is kept open? (The value of E is not specified).

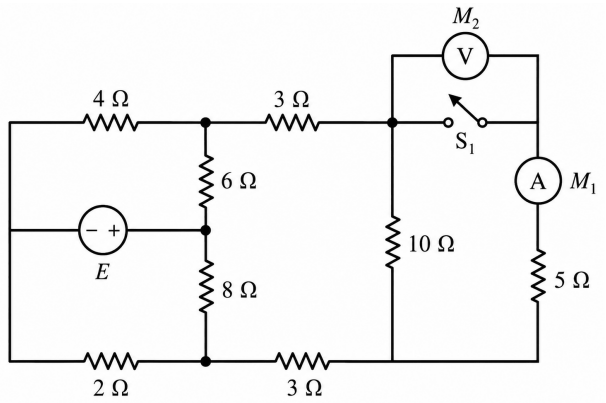


Fig. Q3.93.1

Answer Key

Q3.24.1 2.5	Q3.22.1 B	Q3.20.1 B
Q3.19.1 B	Q3.18.1 7.90 to 8.10	Q3.17.1 -1.01 to -0.99
Q3.16.1 B	Q3.16.2 0.78 to 0.82	Q3.16.3 -1.05 to -0.95
Q3.16.4 D	Q3.15.1 10	Q3.15.2 1.30 to 1.35
Q3.15.3 100	Q3.14.1 D	Q3.14.2 C
Q3.14.3 2.60 to 2.62	Q3.14.4 2.79 to 2.81	Q3.14.5 0.49 to 0.51
Q3.14.6 29.08 to 29.10	Q3.14.7 9 to 11	Q3.13.1 B
Q3.12.1 A	Q3.12.2 A	Q3.11.1 C
Q3.10.1 A	Q3.09.1 C	Q3.08.1 A
Q3.07.1 D	Q3.05.1 B	Q3.05.2 C
Q3.03.1 A	Q3.02.1 A	Q3.01.1 D
Q3.00.1 C	Q3.99.1 C	Q3.98.1 A
Q3.95.1 A, D	Q3.95.2 A	Q3.93.1 50

PrepFusion

UNIT

IV

2nd Order Dynamics and Complex Impedance

Topics

1. Source-Free 2nd Order RLC Circuits
2. Initial Conditions and Sourced RLC Circuits
3. LC Oscillators
4. Equivalent Impedance of AC Circuits

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 2	**MCQ:** 1	**MSQ:** 0	**NAT:** 0
1M: 1	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~0M	**Range:** 0M(–)–0M(–)

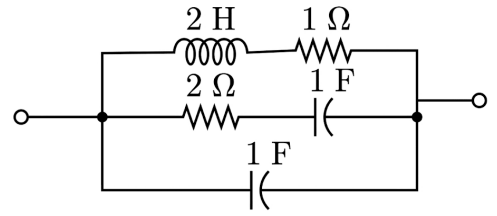
**See preface for how Avg Weightage and Range are calculated.*

Q4.94.1 Descriptive 1994 1M

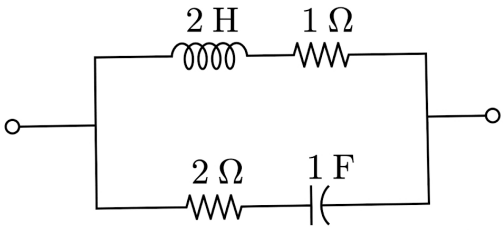
$Z(s) = \dfrac{5}{s^2 + 4}$ represents the input impedance of a network.
(Mention True or False.)

Q4.92.1 MCQ 1992 2M

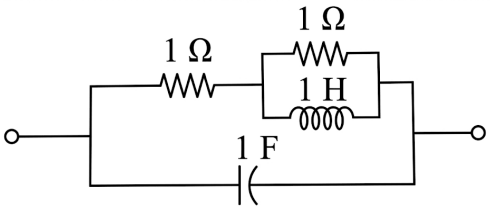
Of the networks N1,N2,N3 and N4 of figure, the networks having identical driving point function are



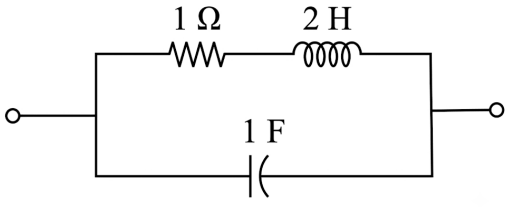
N1



N2



N3



N4

A. N1 and N2
B. N2 and N4
C. N1 and N3
D. N1 and N4

Answer Key

Q4.94.1	Q4.92.1	
False	C	

UNIT

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Introduction to Power in AC Circuits
6. Active, Reactive and Apparent Power in Complex Networks
7. Series RLC Resonance Circuit
8. Quality Factor and Stored Energy in Series Resonance
9. Parallel RLC Resonance Circuit
10. Application of Network Theorems in AC Circuits
11. Maximum Power Transfer in AC Circuits

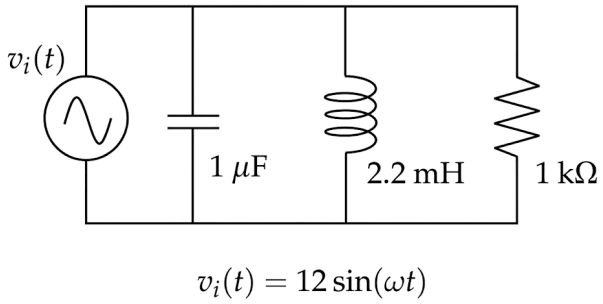
Unit Snapshot*

Stream: EC	**Questions:** 84	**MCQ:** 58	**MSQ:** 1	**NAT:** 24
1M: 39	**2M:** 38	**Descriptive:** 7	**Avg Weightage** ~3M	**Range:** 1M(2023)–5M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 NAT 2026 1M

Consider the circuit shown in the Figure, where the input $v_i(t)$ is in Volt.
The average power (in mW) dissipated in the load resistance of 1 $k\Omega$ at the resonant frequency is _____.
(*rounded off to two decimal places*)



$$v_i(t) = 12\sin(\omega t)$$

Fig. Q5.26.1

Q5.26.2 MCQ 2026 2M

In the given circuit, $L = 1$ μH and $C = 1$ μF. The phasor diagram for I_C and I_L is also shown. Assume that the phase $(\theta_1 + \theta_2)$ is 90° at a frequency of 159.15 kHz. Among the following options, what is the nearest integer value of $R_C \times R_L$?

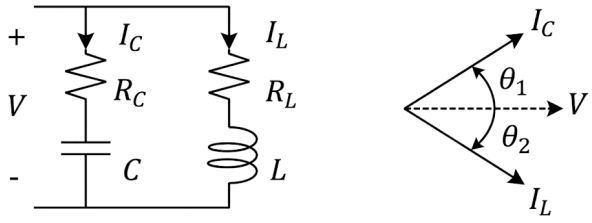


Fig. Q5.26.2

A. 0 **B.** 1 **C.** 2 **D.** 10

Q5.25.1 MCQ 2025 1M

Let i_C,i_L and i_R be the currents flowing through the capacitor, inductor, and resistor, respectively, in the circuit given below. The AC admittances are given in Siemens (S).
Which one of the following is TRUE?

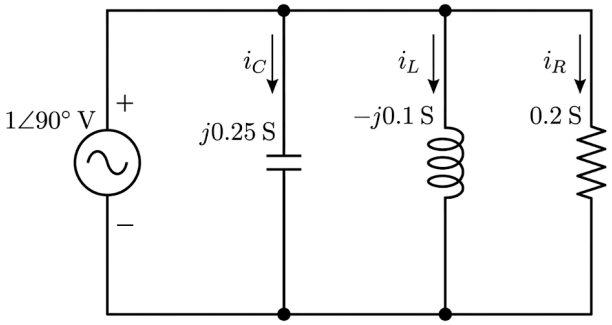


Fig. Q5.25.1

A. $i_C = 0.25\angle 180°$ A, $i_L = 0.1\angle 0°$ A, $i_R = 0.2\angle 90°$ A

B. $i_C = 4\angle 180°$ A, $i_L = 10\angle 0°$ A, $i_R = 5\angle 90°$ A

C. $i_C = 0.25\angle 270°$ A, $i_L = 0.1\angle 90°$ A, $i_R = 0.2\angle 90°$ A

D. $i_C = 4\angle 90°$ A, $i_L = 1\angle 270°$ A, $i_R = 5\angle 0°$ A

Q5.25.2 MCQ 2025 2M

In the circuit below, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100cos(200t)$. What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively?

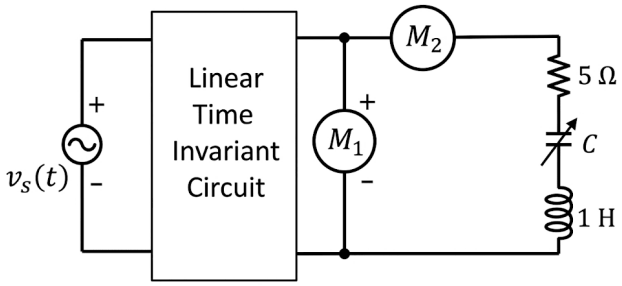


Fig. Q5.25.2

A. 25 μF **B.** 4 μF
C. 0.25 μF **D.** Insufficient information to find C

Q5.23.1 MCQ 2023 1M

A series RLC circuit has a quality factor Q of 1000 at a center frequency of 10^6 rad/s. The possible values of R, L and C are

A. $R = 1\,\Omega, L = 1\,\mu$H, $C = 1\,\mu$F

B. $R = 0.1\,\Omega, L = 1\,\mu$H, $C = 1\,\mu$F

C. $R = 0.01\,\Omega, L = 1\,\mu\text{H}, C = 1\,\mu\text{F}$

D. $R = 0.001\,\Omega, L = 1\,\mu\text{H}, C = 1\,\mu\text{F}$

Q5.22.1 MCQ 2022 2M

For the circuit shown, the locus of the impedance $Z(j\omega)$ is plotted as ω increases from zero to infinity. The values of $R_1$and R_2 are:

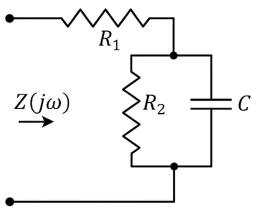


Fig. Q5.22.1

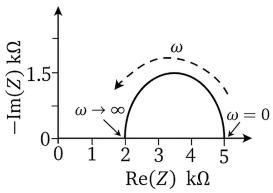


Fig. Q5.22.1

A. $R_1 = 2\text{k}\Omega, R_2 = 3\text{k}\Omega$ **B.** $R_1 = 5\text{k}\Omega, R_2 = 2\text{k}\Omega$
C. $R_1 = 5\text{k}\Omega, R_2 = 2.5\text{k}\Omega$ **D.** $R_1 = 2\text{k}\Omega, R_2 = 5\text{k}\Omega$

Q5.22.2 MCQ 2022 2M

Consider the circuit shown in the figure with input $V(t)$ in volts. The sinusoidal steady state current $I(t)$ flowing through the circuit is shown graphically (where t is in seconds). The circuit element Z can be ___.

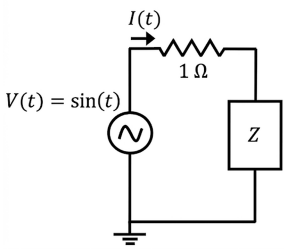


Fig. Q5.22.2

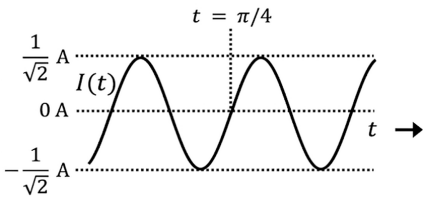


Fig. Q5.22.2

A. a capacitor of $1F$ **B.** an inductor of $1H$
C. a tcapacitor of $\sqrt{3}F$ **D.** an inductor of $\sqrt{3}H$

Q5.20.1 NAT 2020 1M

The current in the R-L circuit shown below is $i(t) = 10cos(5t - \pi/4)A$. The value of the inductor (**rounded off to two decimal places**) is _____ H.

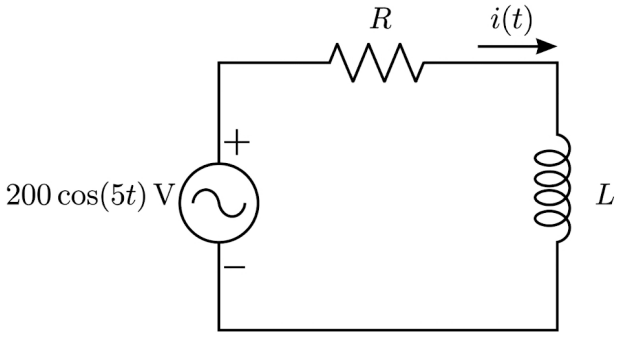


Fig. Q5.20.1

Q5.20.2 MCQ 2020 2M

The current I in the given network is

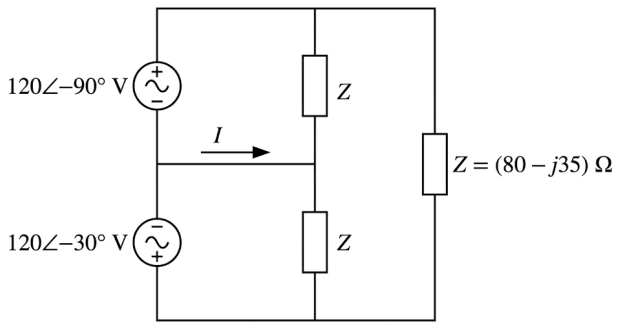


Fig. Q5.20.2

A. 0 A **B.** $2.38 \angle -96.37°$ A
C. $2.38 \angle 143.63°$ A **D.** $2.38 \angle -23.63°$ A

Q5.19.1 MCQ 2019 2M

In the circuit shown, if $v(t) = 2sin(1000t)$ volts, $R = 1\ k\,\Omega$ and $C = 1\ \mu$F, then the steady-state current $i(t)$, in milliamperes (mA), is

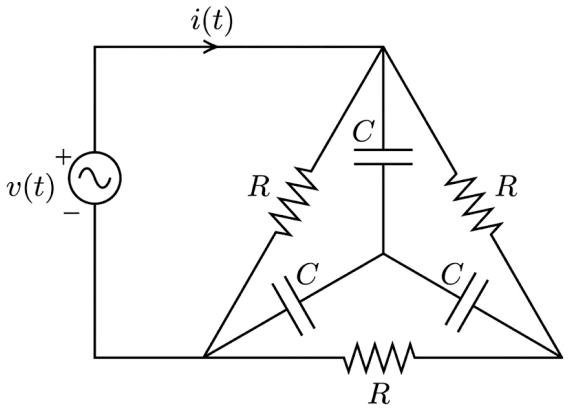


Fig. Q5.19.1

A. $sin(1000t) + cos(1000t)$

B. $2sin(1000t) + 2cos(1000t)$

C. $3sin(1000t) + cos(1000t)$

D. $sin(1000t) + 3cos(1000t)$

Q5.18.1 MCQ 2018 2M

For the circuit given in the figure, the voltage V_C (in volts) across the capacitor is

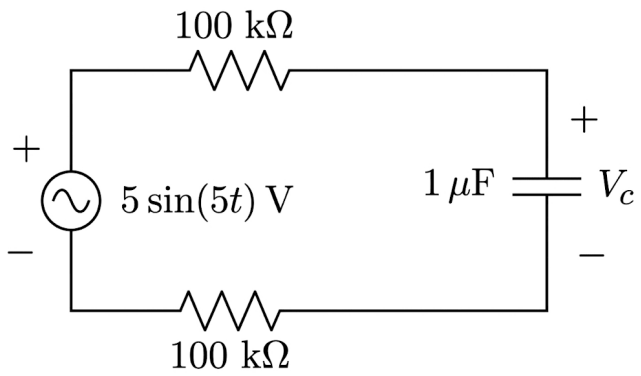


Fig. Q5.18.1

A. $1.25\sqrt{2}sin(5t - 0.25\,\pi)$

B. $1.25\sqrt{2}sin(5t - 0.125\,\pi)$

C. $2.5\sqrt{2}sin(5t - 0.25\,\pi)$

D. $2.5\sqrt{2}sin(5t - 0.125\,\pi)$

Q5.17.1 NAT 2017 2M

The figure shows an RLC circuit excited by the sinusoidal voltage $100\cos(3t)$ Volts, where t is in seconds. The ratio $\frac{\text{amplitude of } V_2}{\text{amplitude of } V_1}$ is _____.

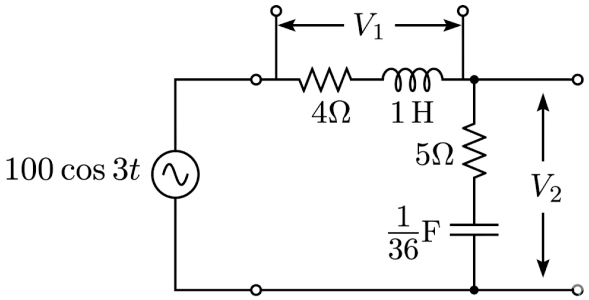


Fig. Q5.17.1

Q5.17.2 NAT 2017 1M

In the circuit shown, the positive angular frequency ω (in radians per second) at which the magnitude of the phase difference between the voltages V_1 and V_2 equals $\frac{\pi}{4}$ radians,is _____.

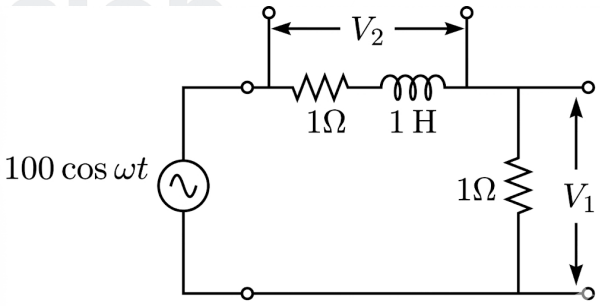


Fig. Q5.17.2

Q5.17.3 NAT 2017 1M

In the circuit shown, V is a sinusoidal voltage source. The current I is in phase with voltage V. The ratio $\frac{\text{amplitude of voltage across the capacitor}}{\text{amplitude of voltage across the resistor}}$ is _____.

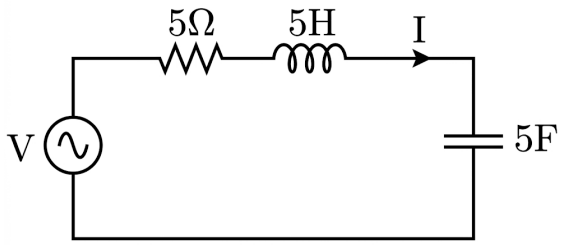


Fig. Q5.17.3

Q5.16.1 NAT 2016 1M

The figure shows an RLC circuit with a sinusoidal current source.

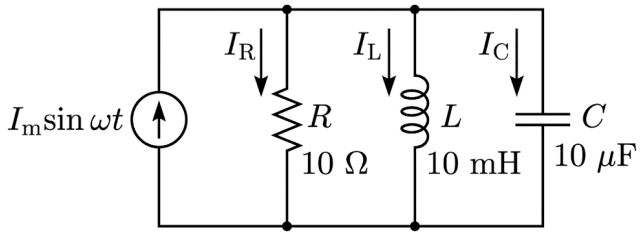


Fig. Q5.16.1

At resonance, the ratio $|\mathbf{I_L}|/|\mathbf{I_R}|$, i.e., the ratio of the magnitudes of the inductor current phasor and the resistor current phasor, is _____

Q5.16.2 MCQ 2016 1M

In the RLC circuit shown in the figure, the input voltage is given by

$$v_i(t) = 2\cos(200t) + 4\sin(500t).$$

The output voltage $v_o(t)$ is

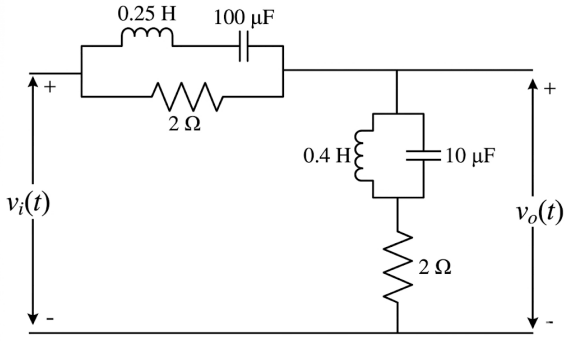


Fig. Q5.16.2

A. $\cos(200t) + 2\sin(500t)$

B. $2\cos(200t) + 4\sin(500t)$

C. $\sin(200t) + 2\cos(500t)$

D. $2\sin(200t) + 4\cos(500t)$

Q5.15.1 NAT 2015 1M

In the circuit shown, at resonance,the amplitude of the sinusoidal voltage (in Volts) across the capacitor is _____.

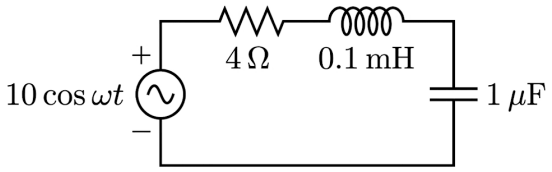


Fig. Q5.15.1

Q5.15.2 NAT 2015 2M

In the given circuit, the maximum power (in Watts) that can be transferred to the load R_L is _____.

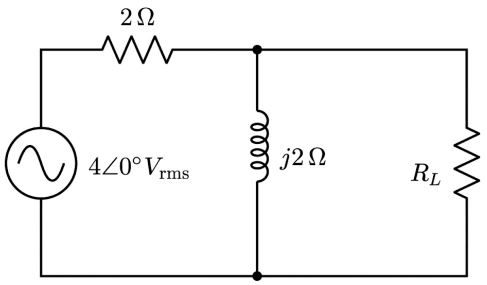


Fig. Q5.15.2

Q5.15.3 NAT 2015 1M

The voltage V_C across the capacitor (in Volts) in the network shown is _____.

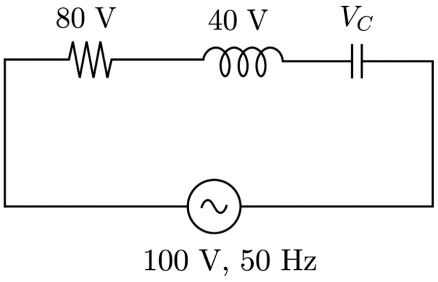


Fig. Q5.15.3

Q5.15.4 NAT 2015 1M

In the circuit shown, the *average* value of the voltage V_{ab} (in Volts) in steady state condition is _____.

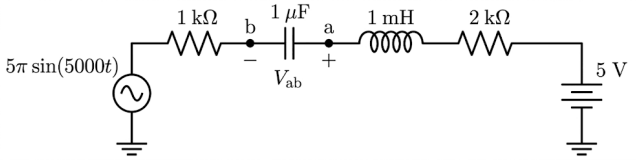


Fig. Q5.15.4

Q5.15.5 NAT 2015 1M

At very high frequencies, the peak output voltage V_0 (in Volts) is _____.

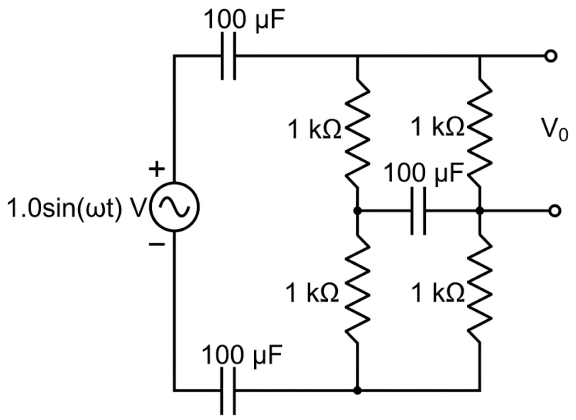


Fig. Q5.15.5

Q5.15.6 NAT 2015 2M

In the circuit shown, the current I flowing through the 50 Ω resistor will be zero if the value of capacitor C (in μF) is _____.

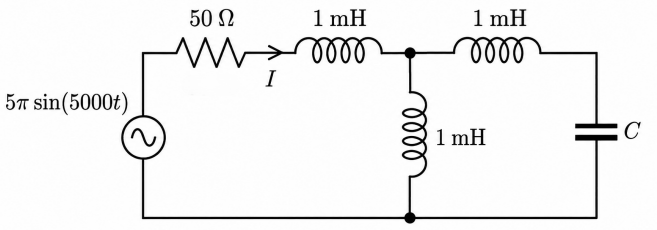


Fig. Q5.15.6

Q5.14.1 MCQ 2014 1M

For maximum power transfer between two cascaded sections of an electrical network, the relationship between the output impedance Z_1 of the first section to the input impedance Z_2 of the second section is

A. $Z_2 = Z_1$ **B.** $Z_2 = -Z_1$ **C.** $Z_2 = Z_1^*$ **D.** $Z_2 = -Z_1^*$

Q5.14.2 MCQ 2014 2M

A 230 V rms source supplies power to two loads connected in parallel. The first load draws 10 kW at 0.8 leading power factor and the second one draws 10 kVA at 0.8 lagging power factor. The complex power delivered by the source is

A. (18 + j 1.5) kVA **B.** (18 – j 1.5) kVA
C. (20 + j 1.5) kVA **D.** (20 – j 1.5) kVA

Q5.14.3 NAT 2014 2M

A series LCR circuit is operated at a frequency different from its resonant frequency. The operating frequency is such that the current leads the supply voltage. The magnitude of current is half the value at resonance. If the values of L, C and R are 1 H, 1 F and 1 Ω, respectively, the operating angular frequency (in rad/s) is _____.

Q5.14.4 MCQ 2014 2M

In the circuit shown in the figure, the value of node voltage V_2 is

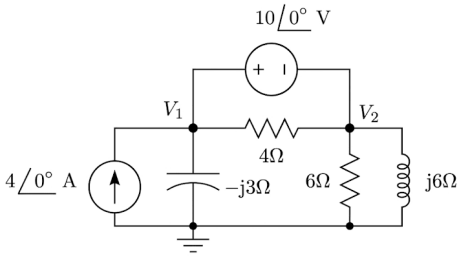


Fig. Q5.14.4

A. 22+j2 V **B.** 2+j22 V
C. 22-j2 V **D.** 2-j22 V

Q5.14.5 NAT 2014 2M

In the circuit shown in the figure, the angular frequency ω (in rad/s), at which the Norton equivalent impedance as seen from the terminals b-b' is purely resistive is _____.

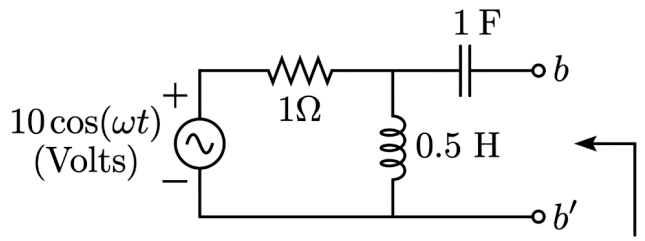


Fig. Q5.14.5

Q5.14.6 MCQ 2014 2M

The steady state output of the circuit shown in the figure is given by $y(t) = A(\omega)sin(\omega t + \phi(\omega)$. If the amplitude $|A(\omega)| = 0.25$,then the frequency ω is

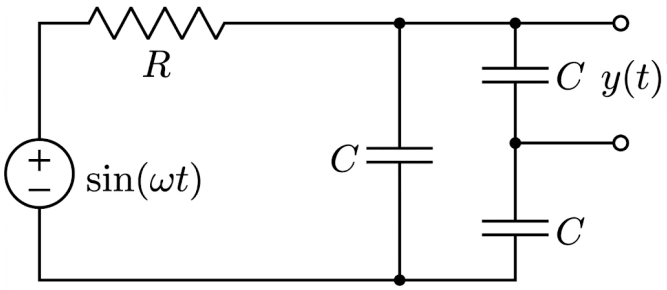


Fig. Q5.14.6

A. $\frac{1}{\sqrt{3}RC}$ **B.** $\frac{2}{\sqrt{3}RC}$ **C.** $\frac{1}{RC}$ **D.** $\frac{2}{RC}$

Q5.13.1 MCQ 2013 1M

A source $V_s(t) = Vcos100\pi t$ has an internal impedance of (4 + j3) Ω. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω should be

A. 3 **B.** 4 **C.** 5 **D.** 7

Q5.13.2 MCQ 2013 2M

In the circuit shown below, if the source voltage $V_s = 100\angle 53.13°$ V then the Thevenin's equivalent voltage in Volts as seen by the load resistance R_L is

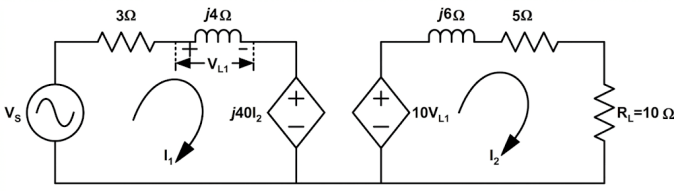


Fig. Q5.13.2

A. $100\angle 90°$ **B.** $800\angle 0°$
C. $800\angle 90°$ **D.** $100\angle 60°$

Q5.13.3 MCQ 2013 2M

The following arrangement consists of an ideal transformer and an attenuator which attenuates by a factor of 0.8. An ac voltage $V_{WX1} = 100$V is applied across WX to get an open circuit voltage V_{YZ1} across YZ. Next, an ac voltage $V_{YZ2} = 100$V is applied across YZ to get an open circuit voltage V_{WX2} across WX. Then, V_{YZ1}/V_{WX1}, V_{WX2}/V_{YZ2} are respectively,

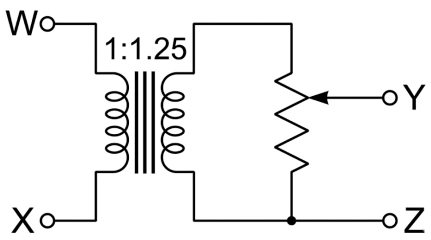


Fig. Q5.13.3

A. 125/100 and 80/100 **B.** 100/100 and 100/100
C. 100/100 and 80/100 **D.** 80/100 and 80/100

Q5.12.1 MCQ 2012 1M

The average power delivered to an impedance (4-j3) Ω by a current $5cos(100\pi t + 100)$ A is

A. 44.2 W **B.** 50 W **C.** 62.5 W **D.** 125 W

Q5.12.2 MCQ 2012 1M

In the circuit shown below, the current through the inductor is

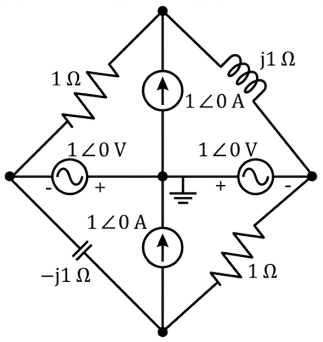


Fig. Q5.12.2

A. $\frac{2}{1+j}$A **B.** $\frac{-1}{1+j}$A **C.** $\frac{1}{1+j}$A **D.** 0 A

Q5.12.3 MCQ 2012 2M

Assuming both the voltage sources are in phase, the value of R for which maximum power is transferred from circuit A to circuit B is

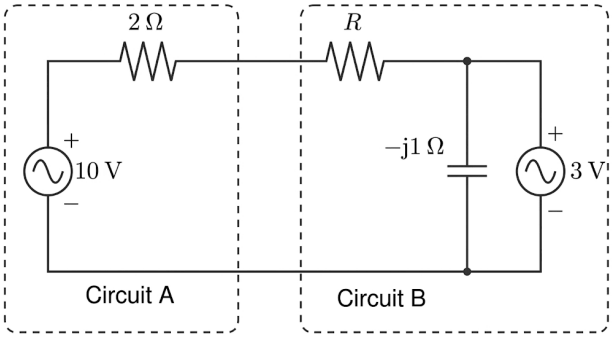


Fig. Q5.12.3

A. 0.8Ω **B.** 1.4Ω **C.** 2Ω **D.** 2.8Ω

Q5.11.1 MCQ 2011 1M

In the circuit shown below, the Norton equivalent current in amperes with respect to the terminals P and Q is

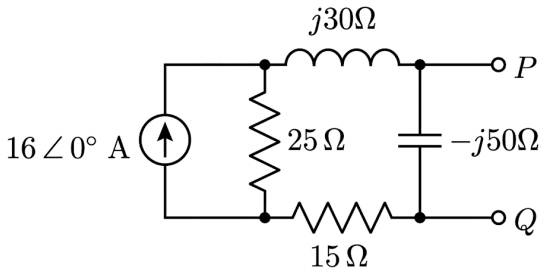


Fig. Q5.11.1

A. 6.4-j4.8 **B.** 6.56-j7.87

C. 10+j0 **D.** 16+j0

Q5.11.2 MCQ 2011 1M

The circuit shown below is driven by a sinusoidal input $v_i = V_p \cos(t/RC)$. The steady state output v_o is

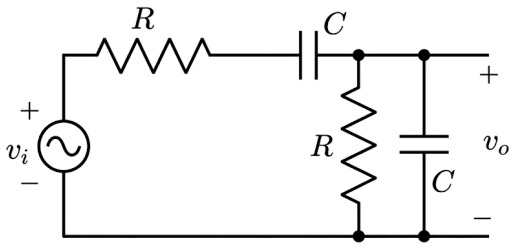


Fig. Q5.11.2

A. $(V_p/3)\cos(t/RC)$ **B.** $(V_p/2)\cos(t/RC)$

C. $(V_p/3)\sin(t/RC)$ **D.** $(V_p/2)\sin(t/RC)$

Q5.11.3 MCQ 2011 2M

In the circuit shown below, the current I is equal to

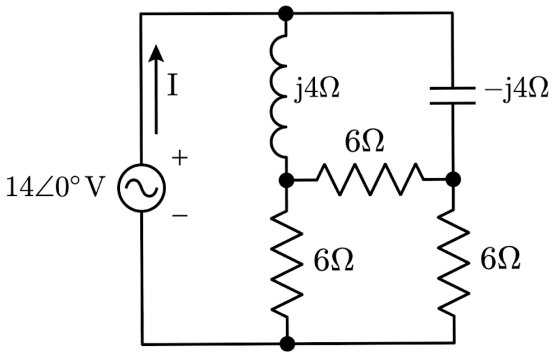


Fig. Q5.11.3

A. $1.4\angle -0°$ A **B.** $2.0\angle 0°$ A

C. $2.8\angle -0°$ A **D.** $3.2\angle -0°$ A

Q5.10.1 MCQ 2010 2M

The current I in the circuit shown is

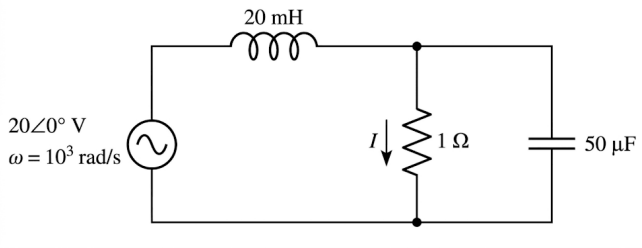


Fig. Q5.10.1

A. -j1 A **B.** j1 A **C.** 0 A **D.** 20 A

Q5.10.2 MCQ 2010 1M

For a parallel RLC circuit which of the following is NOT correct ?

A. The bandwidth of the circuit decreases if R is increased

B. The bandwidth of the circuit remains the same if L is increased

C. At resonance, the input impedance is a real quantity

D. At resonance, the magnitude of input impedance attains its minimum value

Q5.09.1 MCQ 2009 2M

An AC source of RMS voltage 20 V with internal impedance $Z_s = (1+2j)\ \Omega$ feeds a load of impedance $Z_L = (7+4j)\ \Omega$ in the figure below. The reactive power consumed by the load is

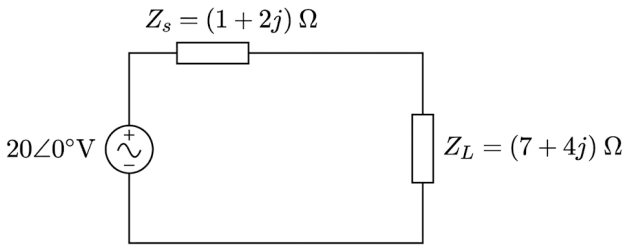


Fig. Q5.09.1

A. 8 VAR **B.** 16 VAR **C.** 28 VAR **D.** 32 VAR

Q5.07.1 MCQ 2007 2M

In the AC network shown in the figure, the phasor voltage V_{AB} (in Volts) is

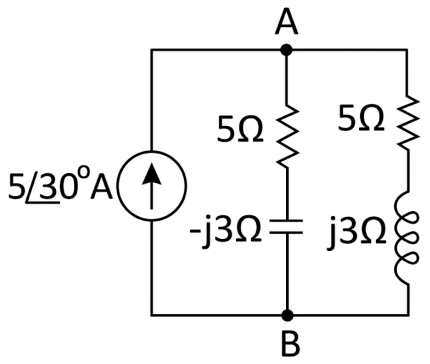


Fig. Q5.07.1

A. 0 **B.** 5∠30° **C.** 12.5∠30° **D.** 17∠30°

Q5.07.2 MCQ 2007 2M

Two series resonant filters are as shown in the figure. Let the 3-dB bandwidth of Filter 1 be B_1 and that of filter 2 be B_2.The value of $\frac{B_1}{B_2}$ is

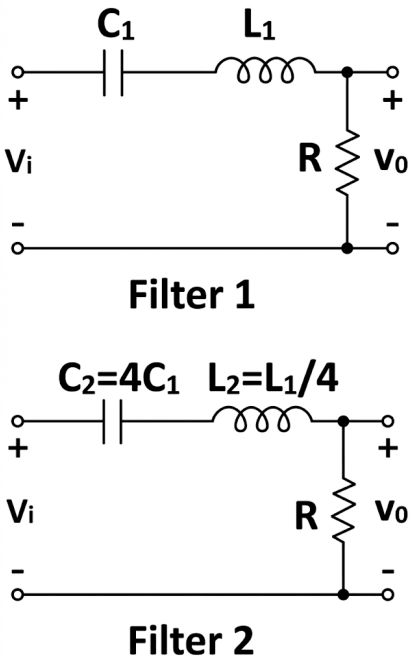


Fig. Q5.07.2

A. 4 **B.** 1 **C.** $\frac{1}{2}$ **D.** $\frac{1}{4}$

Q5.07.3 MCQ 2007 1M

The RC circuit shown in the figure is

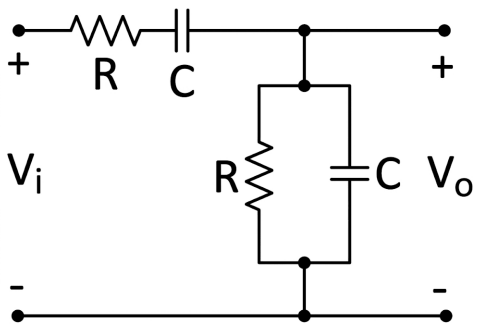


Fig. Q5.07.3

A. A low-pass filter
B. A high-pass filter
C. A band-pass filter
D. A band-reject filter

Q5.07.4 MCQ 2007 1M

An independent voltage source in series with an impedance $Z_S = R_S + jX_S$ delivers a maximum average power to a load impedance Z_L when

A. $Z_L = R_S + jX_S$
B. $Z_L = R_S$
C. $Z_L = jX_S$
D. $Z_L = R_S - jX_S$

Q5.06.1 MCQ 2006 2M

A negative resistance R_{neg} is connected to a passive network N having driving point impedance $Z_1(s)$ as shown below. For $Z_2(s)$ to be positive real,

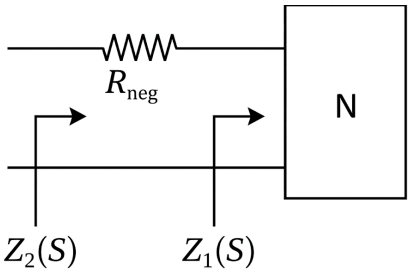


Fig. Q5.06.1

A. $|R_{neg}| \leq ReZ_1(j\omega), \forall\omega$

B. $|R_{neg}| \leq |Z_1(j\omega)|, \forall\omega$

C. $|R_{neg}| \leq ImZ_1(j\omega), \forall\omega$

D. $|R_{neg}| \leq \angle Z_1(j\omega), \forall\omega$

Q5.05.1 MCQ 2005 2M

For the circuit in the figure the instantaneous current $i_1(t)$ is

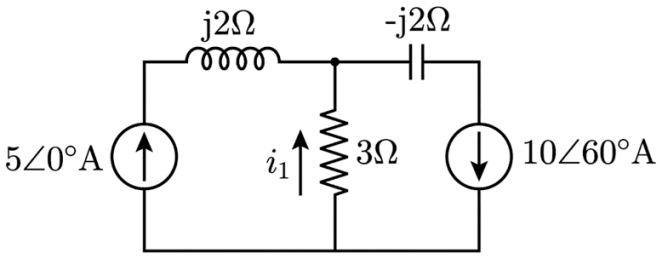


Fig. Q5.05.1

A. $\frac{10\sqrt{3}}{2}\angle 90° Amps$
B. $\frac{10\sqrt{3}}{2}\angle -90° Amps$
C. $5\angle 60° Amps$
D. $5\angle -60° Amps$

Q5.05.2 MCQ 2005 1M

In a series RLC circuit $R = 2\,k\Omega$, $L = 1H$ and $C = \frac{1}{400}\,\mu F$. The resonant frequency is

A. $2 \times 10^4 Hz$
B. $\frac{1}{\pi} \times 10^4 Hz$
C. $10^4 Hz$
D. $2\pi \times 10^4 Hz$

Q5.04.1 MCQ 2004 2M

Consider the following statements S_1 and S_2.
S_1 : At the resonant frequency the impedance of a series R-L-C circuit is zero.
S_2 : In a parallel G-L-C circuit, increasing the conductance G results in increase in its Q factor.
Which one of the folowing is correct ?

A. S_1 is FALSE and S_2 is TRUE.

B. both S_1 and S_2 are TRUE.

C. S_1 is TRUE and S_2 is FALSE.

D. both S_1 and S_2 are FALSE.

Q5.04.2 MCQ 2004 2M

The transfer function $H(s) = \frac{V_o(s)}{V_i(s)}$ of an R-L-C circuit is given by $H(s) = \frac{10^6}{s^2 + 20s + 10^6}$. The Quality factor (Q-factor) of this circuit is

A. 25 **B.** 50 **C.** 100 **D.** 5000

Q5.04.3 MCQ 2004 1M

For the circuit shown in Figure, the time constant RC = 1ms. The input voltage is $v_i(t) = \sqrt{2}sin10^3t$. The output voltage $v_o(t)$ is equal to

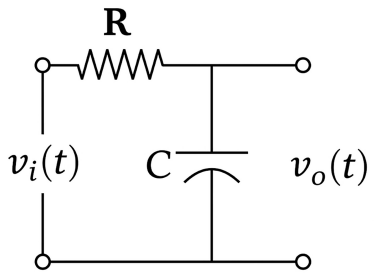


Fig. Q5.04.3

A. $sin(10^3t - 45°)$ **B.** $sin(10^3t + 45°)$

C. $sin(10^3t - 53°)$ **D.** $sin(10^3t + 53°)$

Q5.04.4 MCQ 2004 1M

The circuit shown in figure, with $R = \frac{1}{3}\Omega, L = \frac{1}{4}H, C = 3F$ has input voltage v(t)=sin 2t. The resulting current i(t) is

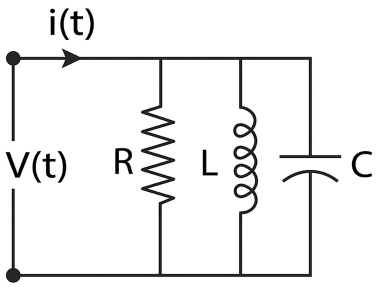


Fig. Q5.04.4

A. $5sin(2t + 53.1°)$ **B.** $5sin(2t - 53.1°)$

C. $25sin(2t + 53.1°)$ **D.** $25sin(2t - 53.1°)$

Q5.03.1 MCQ 2003 1M

A source of angular frequency 1 rad/sec has a source impedance consisting of 1Ω resistance in series with 1 H inductance. The load that will obtain the maximum power transfer is

A. 1Ω resistance

B. 1Ω resistance in parallel with 1 H inductance

C. 1Ω resistance in series with 1 F capacitor

D. 1Ω resistance in parallel with 1 F capacitor

Q5.03.2 MCQ 2003 1M

A series RLC circuit has a resonant frequency of 1 kHz and a quality factor Q = 100. If each R, L and C is doubled from its original value, the new Q of the circuit is

A. 25 **B.** 50 **C.** 100 **D.** 200

Q5.01.1 MCQ 2001 2M

In figure shown below, the value of the load resistor R which maximizes the power delivered to it is

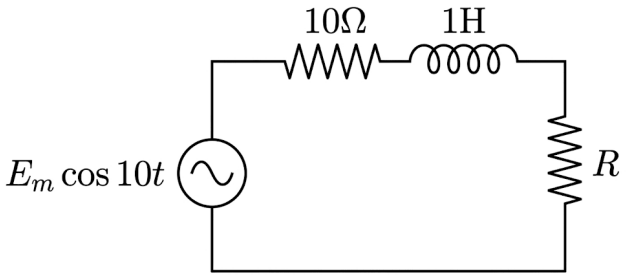


Fig. Q5.01.1

A. 14.14Ω **B.** 10Ω **C.** 200Ω **D.** 28.28Ω

Q5.01.2 MCQ 2001 2M

When the angular frequency ω in figure is varied from 0 to ∞, the locus of the current phasor I_2 is given by

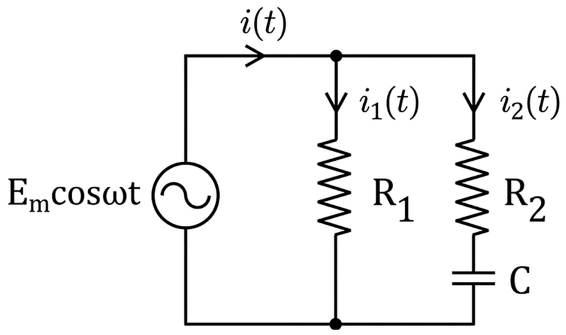
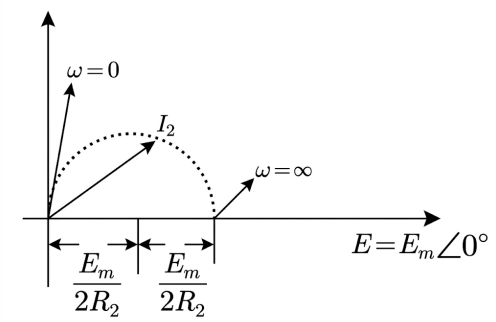
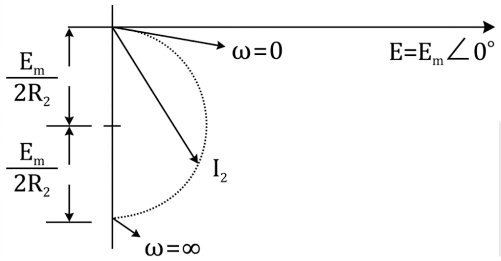


Fig. Q5.01.2

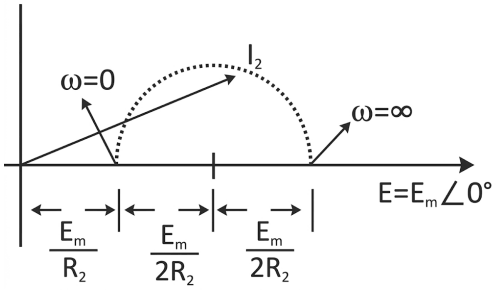
A.



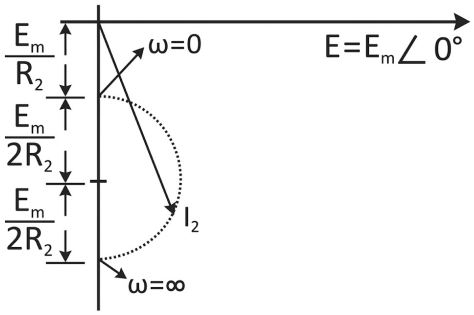
B.



C.



D.



Q5.01.3 NAT 2001 2M

For the circuit shown in Figure, determine the phasors E_2, E_0, I and I_1.

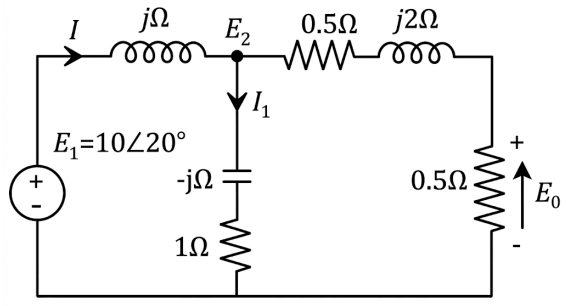


Fig. Q5.01.3

Q5.00.1 MCQ 2000 1M

The circuit of figure represents a

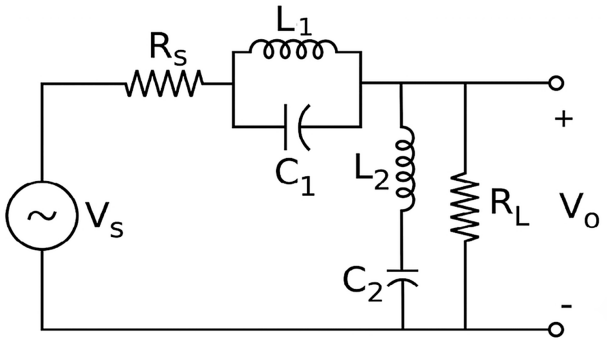


Fig. Q5.00.1

A. Low pass filter
B. High pass filter
C. Band pass filter
D. Band reject filter

Q5.00.2 MCQ 2000 2M

In Figure, the steady state output voltage corresponding to the input voltage 3+ 4 sin 100t V is

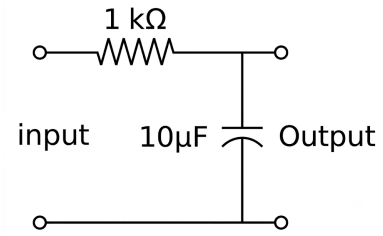


Fig. Q5.00.2

A. $3 + \frac{4}{\sqrt{2}} \sin(100t - \frac{\pi}{4})$ V

B. $3 + 4\sqrt{2} \sin(100t - \frac{\pi}{4})$ V

C. $\frac{3}{2} + \frac{4}{\sqrt{2}} \sin(100t + \frac{\pi}{4})$ V

D. $3 + 4\sin(100t - \frac{\pi}{4})$ V

Q5.00.3 NAT 2000 5M

For the circuit in Figure, which is in steady state

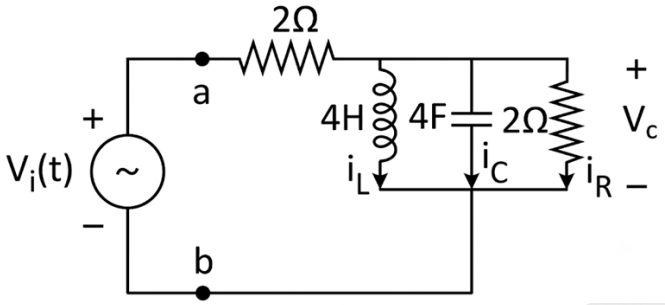


Fig. Q5.00.3

(a) Find the frequency ω_o at which the magnitude of the impedance across terminals a and b reaches a maximum.

(b)Find the impedance across a,b at the frequency ω_o.

(c) If $v_i(t) = V\sin(\omega_o t)$, find $i_L(t)$, $i_C(t)$ and $i_R(t)$

Q5.99.1 MCQ 1999 2M

The Thevenin's equivalent voltage V_{TH} appearing between the terminals A and B of the network shown in Figure is given by

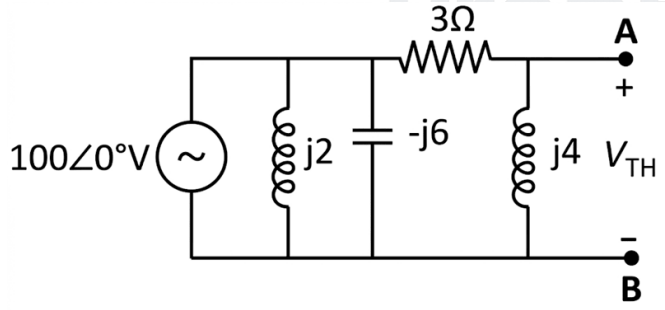


Fig. Q5.99.1

A. j16(3-j4)
B. j16(3+j4)
C. 16(3+j4)
D. 16(3-j4)

Q5.99.2 NAT 1999 5M

For the network shown in Figure, evaluate the current I flowing through the 2Ω resistor using superposition theorem.

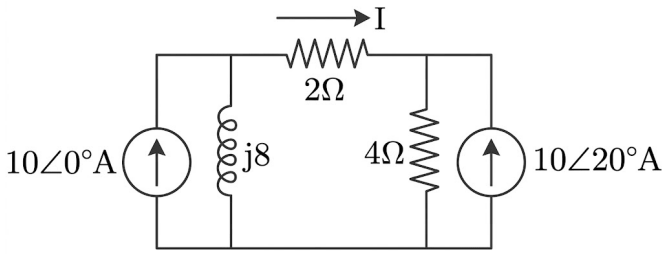


Fig. Q5.99.2

Q5.99.3 NAT 1999 5M

A coil with a quality factor (Q) of 10 is put in series with capacitor C_1 of $10\mu F$, and the combination is found to draw maximum current when a sinusoidal voltage of frequency 50 Hz is applied. A second capacitor C_2 is now in parallel with the circuit. What should be the capacitance of C_2 for combined circuit to act purely as a resistance for a sinusoidal excitation at a frequency of 100 Hz ? Calculate the RMS current drawn by the combined circuit at 100 Hz if the applied voltage is 100 V(RMS).

Q5.98.1 MCQ 1998 1M

The parallel RLC circuit shown in Figure is in resonance. In this circuit

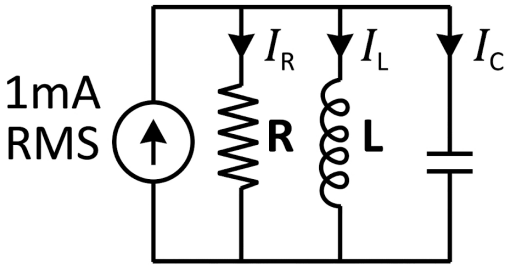


Fig. Q5.98.1

A. $|I_R| < 1\,mA$
B. $|I_R + I_L| > 1\,mA$
C. $|I_R + I_C| < 1\,mA$
D. $|I_R + I_C| > 1\,mA$

Note: For the printed ideal parallel-resonant circuit, both options (B) and (D) satisfy the stated condition.

Q5.98.2 NAT 1998 3M

Determine the frequency of resonance and the resonant impedance of the parallel circuit shown in the figure. What happens when $L = CR^2$?

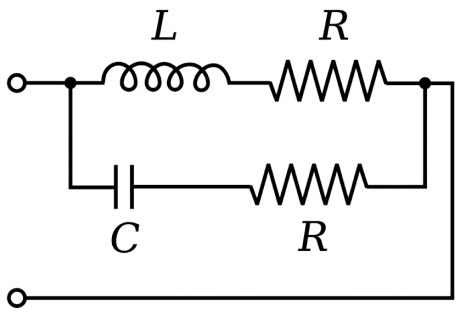


Fig. Q5.98.2

Q5.98.3 NAT 1998 3M

A voltage source of internal impedance $R_S + jX_S$ supplies power to a load of impedance $R_L + jX_L$ in which only R_L is variable. Determine the value of R_L foe maximum power transfer from the source to the load. Aso, find the numerical value of R_L if the source impedance is 3.0Ω (purely resistive) and X_1 is 4.0Ω.

Q5.97.1 MCQ 1997 1M

In the circuit of the figure is the equivalent impedance seen across terminals a, b is

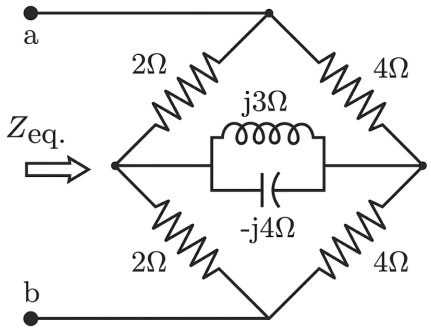


Fig. Q5.97.1

A. $(\frac{16}{3})\Omega$ **B.** $(\frac{8}{3})\Omega$

C. $(\frac{8}{3} + 12j)\Omega$ **D.** None of the above

Q5.97.2 NAT 1997 5M

In the circuit of the figure all currents and voltages are sinusoids of frequency ω rad/sec.

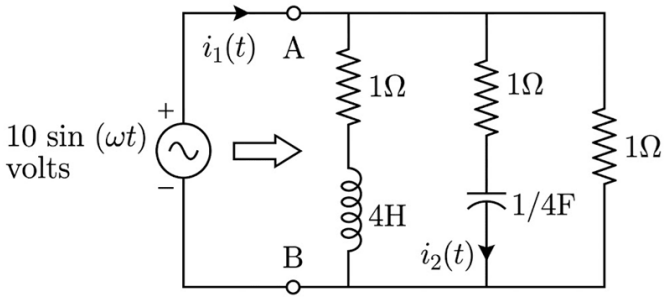


Fig. Q5.97.2

(a) Find the impedance to the right of (A,B) at $\omega = 0$ rad/sec and $\omega = \infty$ rad/sec.

(b) If $\omega = \omega_0$ rad/sec (resonance frequency) and $i_1(t) = I \sin \omega_0 t$ A, where I is positive, $\omega_0 \neq 0, \omega_0 \neq \infty$, then find I, ω_0 and $i_2(t)$.

Q5.97.3 NAT 1997 5M

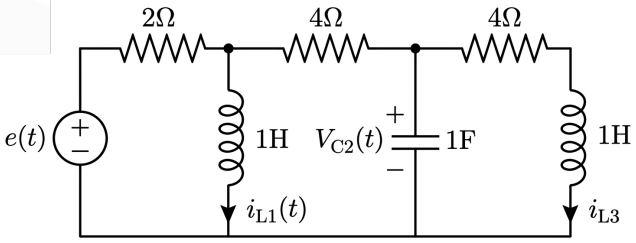


Fig. Q5.97.3

For the circuit shown in the figure is choose state variables X_1, X_2, X_3, to be $i_{L1}(t), V_{c2}(t), i_{L3}(t)$

(a) Write the state equations

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \dot{X}_3 \end{bmatrix} = A \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + B \left[e(t) \right]$$

(b) If $e(t) = 0, t \geq 0, i_{L1}(0) = 0, V_{c2}(0) = 0, i_{L3}(0) = 1\text{A}$, then what would the total energy dissipated in the resistors in the interval $(0, \infty)$ be?

Q5.96.1 MCQ 1996 1M

In the given figure A_1, A_2 and A_3 are ideal ammeters. If A_2 and A_3 read 3A and 4A, respectively, then A_1 should read

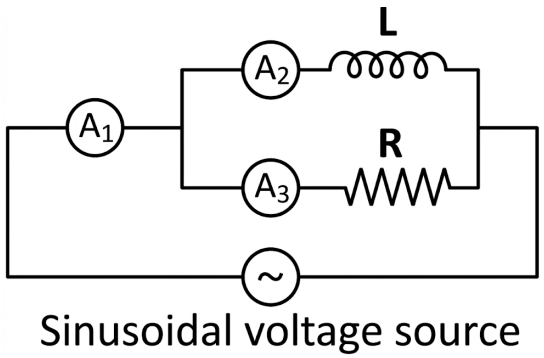


Fig. Q5.96.1

A. 1A **B.** 5A **C.** 7A **D.** None of these

Q5.95.1 MCQ 1995 1M

The current i(t), through a 10Ω resistor in series with an inductance is given by

$$i(t) = 3 + 4\sin(100t + 45°) + 4\sin(300t + 60°)A$$

The RMS value of the current and the power dissipated in the circuit are :

A. $\sqrt{41}A, 410W$ respectively

B. $\sqrt{35}A, 350W$ respectively

C. $5A, 250W$ respectively

D. $11A, 1210W$ respectively

Q5.95.2 Descriptive 1995 1M

A series R-L-C circuit has a Q of 100 and an impedance of (100+j0)Ω at its resonant angular frequency of 10^7 rad/sec. The values of R and L are :

R = _____ ohms and L = _____ ohms.

Q5.95.3 NAT 1995 2M

Two 2H inductance coils are connected in series and are also magnetically coupled to each other the coefficient of coupling being 0.1. The total inductance of the combination can be

Q5.94.1 MCQ 1994 1M

A series RLC circuit consisting of $R = 10\Omega$, $|X_L| = 20\Omega$ and $|X_C| = 20\Omega$, is connected across an a.c. supply of 200 V rms. The rms voltage across the capacitor is

A. $200\angle -90°$ **B.** $200\angle +90°$

C. $400\angle +90°$ **D.** $400\angle -90°$

Q5.93.1 MCQ 1993 1M

In the series circuit shown in figure for series resonance, the value of the coupling coefficient K will be

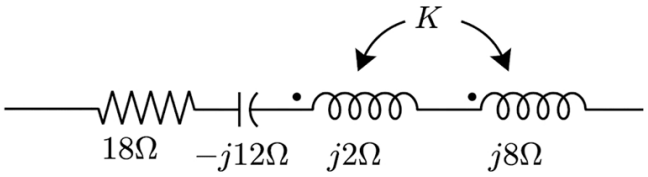


Fig. Q5.93.1

A. 0.25 **B.** 0.5 **C.** 0.999 **D.** 1.0

Q5.93.2 MCQ 1993 2M

If the secondary winding of the ideal transformer shown in the circuit of figure has 40 turns, the number of turns in the primary winding for maximum power transfer to the 2Ω resistor will be

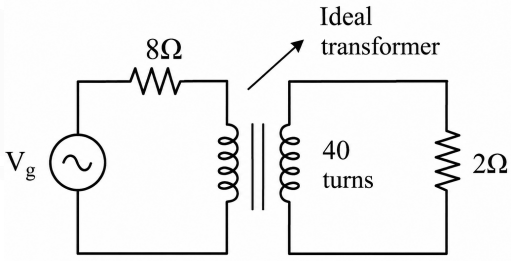


Fig. Q5.93.2

A. 20 **B.** 40 **C.** 80 **D.** 160

Q5.93.3 MCQ 1993 1M

In figure, A_1, A_2 and A_3 are ideal ammeters. If A_1 reads 5A, A_2 reads 12A , then A_3 should read

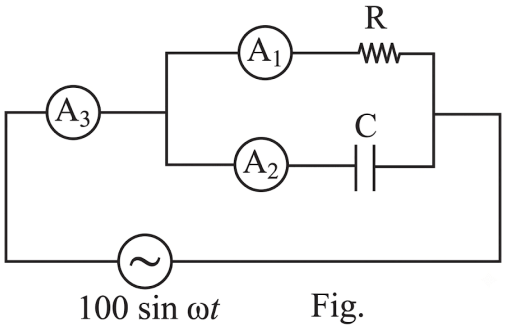


Fig. Q5.93.3

A. 7A **B.** 12A **C.** 13A **D.** 17A

Q5.92.1 MCQ 1992 2M

For the series R-L-C circuit of figure, the partial phasor diagram at a certain frequency is shown in figure. The operating frequency of the circuit is :

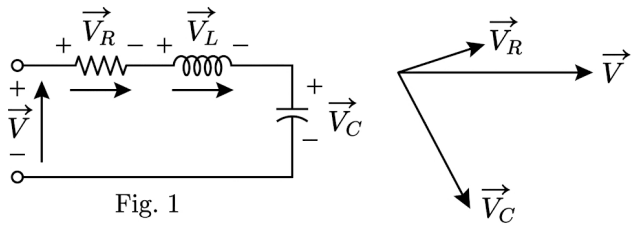


Fig. 1

Fig. 2

Fig. Q5.92.1

A. Equal to the resonance frequency

B. Lesser than the resonance frequency

C. Greater than the resonance frequency

D. Not zero

Q5.91.1 MCQ 1991 1M

In a series RLC high Q circuit, the current peaks at a frequency

A. Equal to the resonant frequency

B. Lesser than the resonant frequency

C. Greater than the resonant frequency

D. None of the above

Q5.90.1 MCQ 1990 1M

The resonant frequency of the series circuit shown in figure is

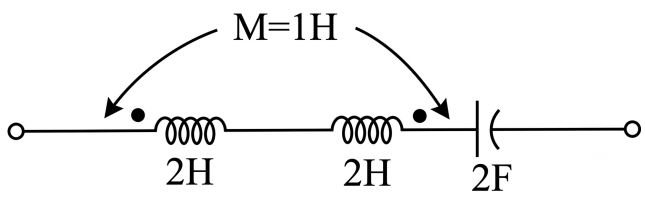


Fig. Q5.90.1

A. $\frac{1}{4\pi\sqrt{3}}Hz$ **B.** $\frac{1}{4\pi}Hz$

C. $\frac{1}{2\pi\sqrt{10}}Hz$ **D.** $\frac{1}{4\pi\sqrt{2}}Hz$

Q5.89.1 MCQ 1989 2M

In the circuit of figure, the power dissipated in the resistor R is 1W when only source '1' is present and '2' is replaced by a short. The power dissipated in the same resistor R is 4W when only source '2' is present and '1' is replaced by a short. When both the sources '1' and '2' are present, the power dissipated in R will be :

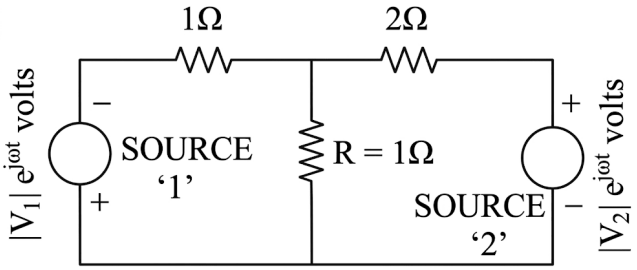


Fig. Q5.89.1

A. 1W **B.** 3W **C.** 4W **D.** 5W

Q5.89.2 MCQ 1989 1M

A load $Z_L = R_L + jX_L$ is to be matched, using an ideal transformer to a generator of internal impedance, $Z_S = R_S + jX_S$. The turns ratio of the transformer required is

A. $\sqrt{|Z_L/Z_S|}$ **B.** $\sqrt{|R_L/R_S|}$

C. $\sqrt{|R_L/Z_S|}$ **D.** $\sqrt{|Z_L/R_S|}$

Q5.89.3 MSQ 1989 2M

The half-power bandwidth of the resonant circuit of figure can be increased by

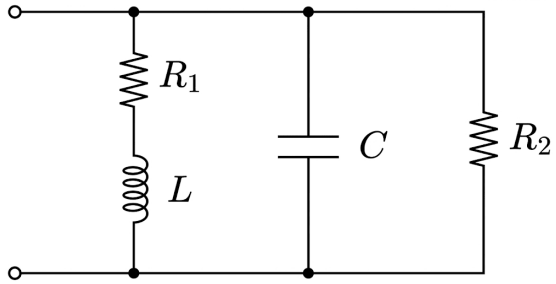


Fig. Q5.89.3

☐ **A.** increasing R_1

☐ **B.** decreasing R_1

☐ **C.** increasing R_2

☐ **D.** decreasing R_2

Q5.88.1 MCQ 1988 1M ○○○

If an impedance Z_L is connected across a voltage source V with source impedance Z_S, then for maximum power transfer, the load impedance must be equal to

A. Source impedance Z_S

B. Complex conjugate of Z_S

C. Real part of Z_S

D. Imaginary part of Z_S

Q5.87.1 NAT 1987 2M ○○○

The value of current through the 1F capacitor of figure is

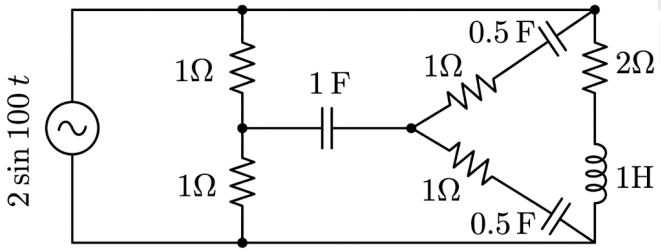


Fig. Q5.87.1

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q5.26.1 71.5 to 72.5	Q5.26.2 B	Q5.25.1 A
Q5.25.2 A	Q5.23.1 D	Q5.22.1 A
Q5.22.2 B	Q5.20.1 2.80 to 2.85	Q5.20.2 C
Q5.19.1 C	Q5.18.1 C	Q5.17.1 2.55 to 2.65
Q5.17.2 0.9 to 1.1	Q5.17.3 0.19 to 0.21	Q5.16.1 0.30 to 0.34
Q5.16.2 B	Q5.15.1 24 to 26	Q5.15.2 1.6 to 1.7
Q5.15.3 100	Q5.15.4 4.9 to 5.1	Q5.15.5 0.49 to 0.51
Q5.15.6 20	Q5.14.1 C	Q5.14.2 B
Q5.14.3 0.45 to 0.47	Q5.14.4 D	Q5.14.5 1.9 to 2.1
Q5.14.6 B	Q5.13.1 C	Q5.13.2 C
Q5.13.3 B	Q5.12.1 B	Q5.12.2 C
Q5.12.3 A	Q5.11.1 A	Q5.11.2 A
Q5.11.3 B	Q5.10.1 A	Q5.10.2 D
Q5.09.1 B	Q5.07.1 D	Q5.07.2 D
Q5.07.3 C	Q5.07.4 D	Q5.06.1 A
Q5.05.1 A	Q5.05.2 B	Q5.04.1 D
Q5.04.2 B	Q5.04.3 A	Q5.04.4 A
Q5.03.1 C	Q5.03.2 B	Q5.01.1 A
Q5.01.2 A	Q5.01.3 *	Q5.00.1 D
Q5.00.2 A	Q5.00.3 *	Q5.99.1 A
Q5.99.2 *	Q5.99.3 *	Q5.98.1 B
Q5.98.2 *	Q5.98.3 5	Q5.97.1 B
Q5.97.2 *	Q5.97.3 *	Q5.96.1 B
Q5.95.1 C	Q5.95.2 100,0.001	Q5.95.3 *
Q5.94.1 D	Q5.93.1 A	Q5.93.2 C
Q5.93.3 C	Q5.92.1 B	Q5.91.1 A
Q5.90.1 B	Q5.89.1 A	Q5.89.2 A
Q5.89.3 A, D	Q5.88.1 B	Q5.87.1 0

UNIT

VI

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 41	**MCQ:** 28	**MSQ:** 2	**NAT:** 11
1M: 15	**2M:** 24	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2020)–4M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 2M

Consider the two-port network as shown in the Figure. Which of the following options provides the correct set of values of A, B, C and D parameters?

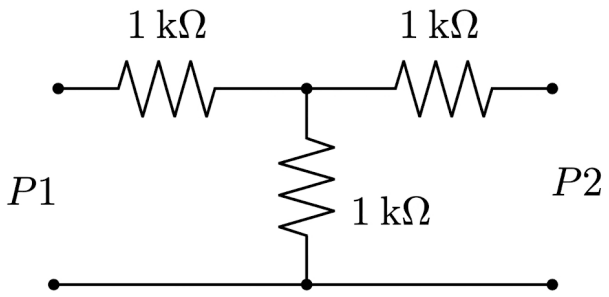


Fig. Q6.26.1

A. $A = 1/2, B = 3 \times 10^3\ \Omega, C = 10^{-3}\ \Omega^{-1}, D = 1/2$

B. $A = 2, B = 3\ \Omega, C = 10^{-3}\ \Omega^{-1}, D = 2$

C. $A = 2, B = 6 \times 10^3\ \Omega, C = 2 \times 10^{-3}\ \Omega^{-1}, D = 2$

D. $A = 2, B = 3 \times 10^3\ \Omega, C = 10^{-3}\ \Omega^{-1}, D = 2$

Q6.25.1 MCQ 2025 2M

The Z-parameter matrix of a two port network relates the port voltages and port currents as follows:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = Z \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

The Z-parameter matrix (with each entry in Ohms) of the network shown below is _____.

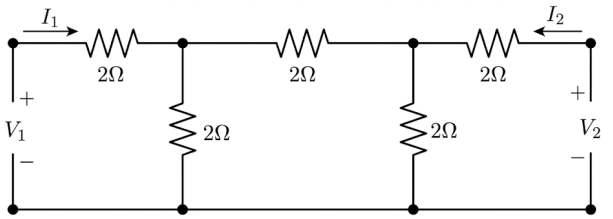


Fig. Q6.25.1

A. $\begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$

B. $\begin{bmatrix} 2/3 & 10/3 \\ 10/3 & 2/3 \end{bmatrix}$

C. $\begin{bmatrix} 10 & 2 \\ 2 & 10 \end{bmatrix}$

D. $\begin{bmatrix} 10/3 & 1/3 \\ 1/3 & 10/3 \end{bmatrix}$

Q6.24.1 NAT 2024 2M

For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____.

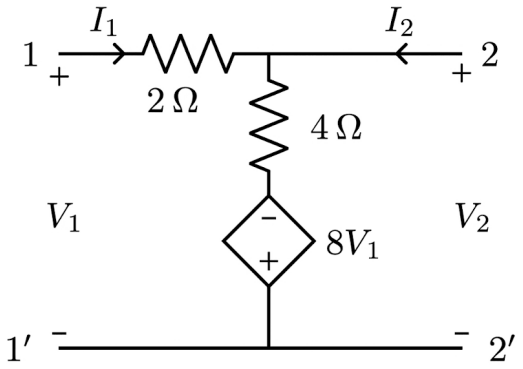


Fig. Q6.24.1

Q6.23.1 NAT 2023 1M

For the two port network shown below, the [Y]-parameters is given as

$$[Y] = \frac{1}{100}\begin{bmatrix} 2 & -1 \\ -1 & 4/3 \end{bmatrix} S$$

The value of load impedance Z_L, in Ω, for maximum power transfer will be_____ (*rounded off to two decimal places*).

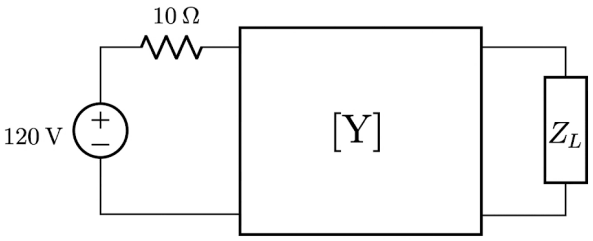


Fig. Q6.23.1

Q6.23.2 MCQ 2023 2M

The h-parameters of a two port network are shown below. The condition for the maximum small signal voltage gain $\frac{Vout}{Vs}$ is

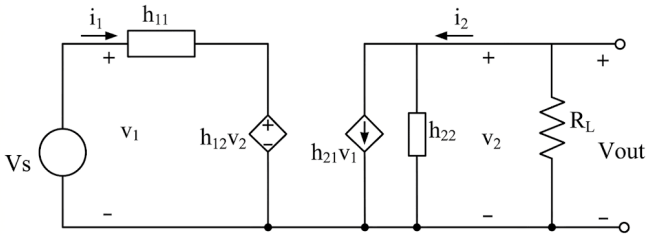


Fig. Q6.23.2

A. $h_{11} = 0, h_{12} = 0, h_{21} =$ very high and $h_{22} = 0$

B. $h_{11} =$ very high, $h_{12} = 0, h_{21} =$ very high and $h_{22} = 0$

C. $h_{11} = 0, h_{12} =$ very high, $h_{21} =$ very high and $h_{22} = 0$

D. $h_{11} = 0, h_{12} = 0, h_{21} =$ very high and $h_{22} =$ very high

Q6.22.1 NAT 2022 2M

A linear 2-port network is shown in Fig. (a). An ideal DC voltage source of 10 V is connected across Port 1. A variable resistance R is connected across Port 2. As R is varied, the measured voltage and current at Port 2 is shown in Fig. (b) as a V_2 versus $-I_2$ plot. Note that for $V_2 = 5V$, $I_2 = 0mA$, and for $V_2 = 4V$, $I_2 = -4mA$.
When the variable resistance R at Port 2 is replaced by the load shown in Fig. (c), the current I_2 is _____ mA (*rounded off to one decimal place*).

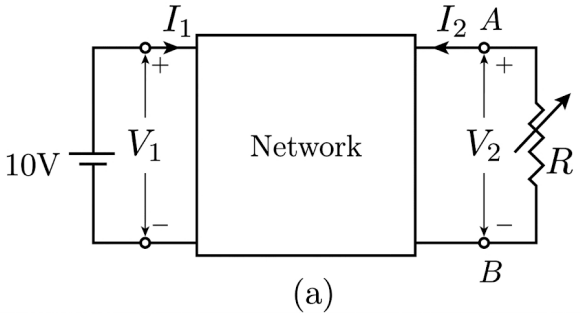


Fig. Q6.22.1

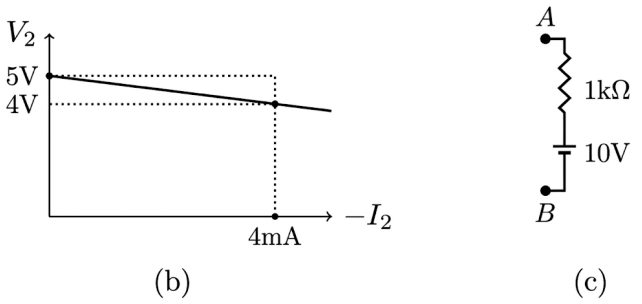


Fig. Q6.22.1

Q6.21.1 MCQ 2021 2M

Consider the two-port network shown in the figure.

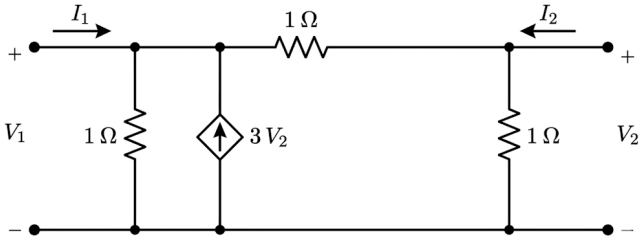


Fig. Q6.21.1

The admittance parameters, in siemens, are

A. $y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 2$

B. $y_{11} = 1, y_{12} = -2, y_{21} = -1, y_{22} = 3$

C. $y_{11} = 2, y_{12} = -4, y_{21} = -1, y_{22} = 2$

D. $y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 3$

Q6.20.1 NAT 2020 1M

In the given circuit, the two-port network has the impedance matrix $[Z] = \begin{bmatrix} 40 & 60 \\ 60 & 120 \end{bmatrix}$. The value of Z_L, in Ω, for which maximum power is transferred to the load is _____ Ω.

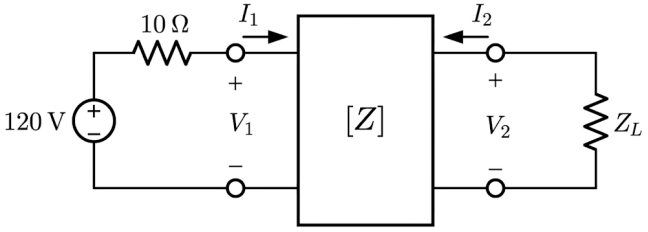


Fig. Q6.20.1

Q6.18.1 NAT 2018 1M

The $ABCD$ matrix for a two-port network is defined by :

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

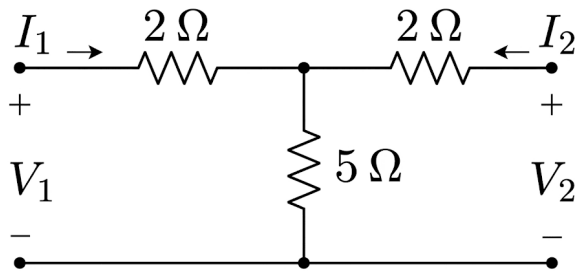


Fig. Q6.18.1

The parameter B for the two-port network (in ohms, correct to two decimal places) is _____.

Q6.16.1 MCQ 2016 1M

Consider a two-port network with the transmission matrix: $T = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$.If the network is reciprocal, then

A. $T^{-1} = T$ **B.** $T^2 = T$

C. Determinant$(T) = 0$ **D.** Determinant$(T) = 1$

Q6.16.2 NAT 2016 1M

The z-parameter matrix for the two-port network shown is

$$\begin{bmatrix} 2j\omega & j\omega \\ j\omega & 3+2j\omega \end{bmatrix}$$

where the entries are in Ω. Suppose $Z_b(j\omega) = R_b + j\omega$.

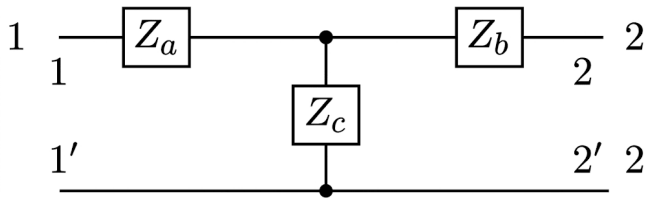


Fig. Q6.16.2

Then the value of R_b (in Ω) equals _____ .

Q6.16.3 MCQ 2016 2M

The z-parameter matrix $\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$ for the two-port network shown is

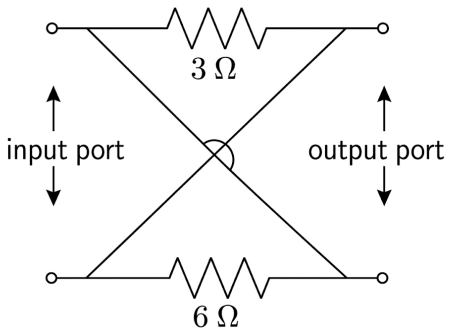


Fig. Q6.16.3

A. $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ **B.** $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$

C. $\begin{bmatrix} 9 & -3 \\ 6 & 9 \end{bmatrix}$ **D.** $\begin{bmatrix} 9 & 3 \\ 6 & 9 \end{bmatrix}$

Q6.15.1 MCQ 2015 1M

The 2-port admittance matrix of the circuit shown is given by

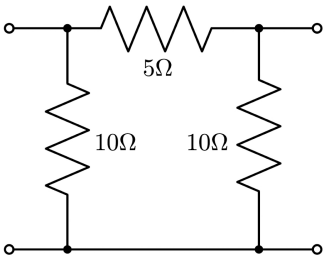


Fig. Q6.15.1

A. $\begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$ **B.** $\begin{bmatrix} 15 & 5 \\ 5 & 15 \end{bmatrix}$

C. $\begin{bmatrix} 3.33 & 5 \\ 5 & 3.33 \end{bmatrix}$ **D.** $\begin{bmatrix} 0.3 & 0.4 \\ 0.4 & 0.3 \end{bmatrix}$

Q6.15.2 MCQ 2015 2M

The ABCD parameters of the following 2-port network are

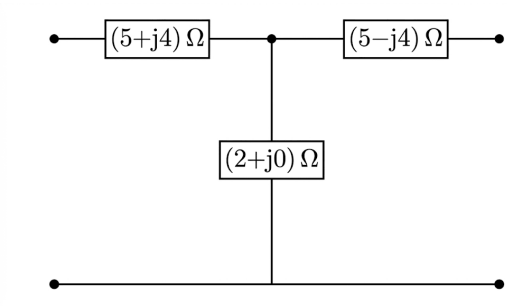


Fig. Q6.15.2

A. $\begin{bmatrix} 3.5+j2 & 20.5 \\ 20.5 & 3.5-j2 \end{bmatrix}$ **B.** $\begin{bmatrix} 3.5+j2 & 30.5 \\ 0.5 & 3.5-j2 \end{bmatrix}$

C. $\begin{bmatrix} 10 & 2+j0 \\ 2+j0 & 10 \end{bmatrix}$ **D.** $\begin{bmatrix} 7+j4 & 0.5 \\ 30.5 & 7-j4 \end{bmatrix}$

Q6.14.1 NAT 2014 2M

In the h-parameter model of the 2-port network given in the figure shown, the value of h_{22} (in S), is _____.

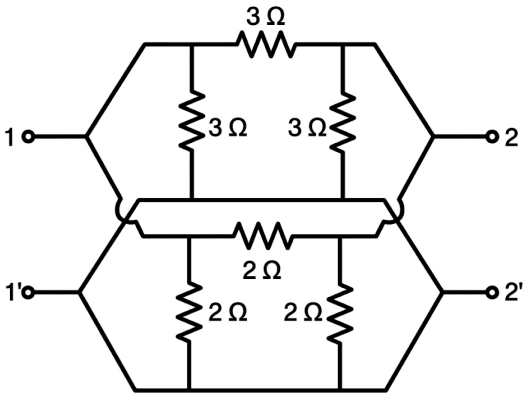


Fig. Q6.14.1

Q6.14.2 MCQ 2014 2M

For the two-port network shown in the figure, the impedance (Z) matrix (in Ω) is

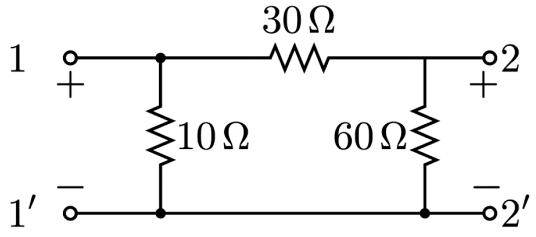


Fig. Q6.14.2

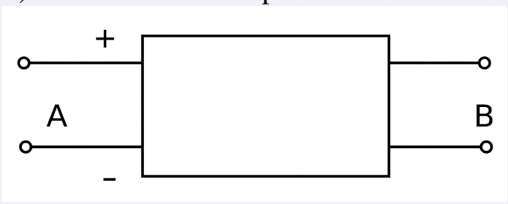
A. $\begin{bmatrix} 6 & 24 \\ 42 & 9 \end{bmatrix}$ **B.** $\begin{bmatrix} 9 & 8 \\ 8 & 24 \end{bmatrix}$

C. $\begin{bmatrix} 9 & 6 \\ 6 & 24 \end{bmatrix}$ **D.** $\begin{bmatrix} 42 & 6 \\ 46 & 60 \end{bmatrix}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q6.12.1 and Q6.12.2 using the following data. With 10 V dc connected at port A in the linear nonreciprocal two-port network shown below, the following were observed:
i) 1 Ω connected at port B draws a current of 3 A
ii) 2.5 Ω connected at port B draws a current of 2 A



Q6.12.1 MCQ 2012 2M

With 10 V dc connected at port A, the current drawn by 7 Ω connected at port B is

A. 3/7A **B.** 5/7A **C.** 1A **D.** 9/7A

Q6.12.2 MCQ 2012 2M

For the same network, with 6 V dc connected at port A, 1 Ω connected at port B draws 7/3 A. If 8 V dc is connected to port A, the open circuit voltage at port B is

A. 6V **B.** 7V **C.** 8V **D.** 9V

Q6.11.1 MCQ 2011 2M

In the circuit shown below, the network N is described by the following Y matrix:

$$Y = \begin{bmatrix} 0.1\,\text{S} & -0.01\,\text{S} \\ 0.01\,\text{S} & 0.1\,\text{S} \end{bmatrix}$$

The voltage gain $\frac{V_2}{V_1}$ is

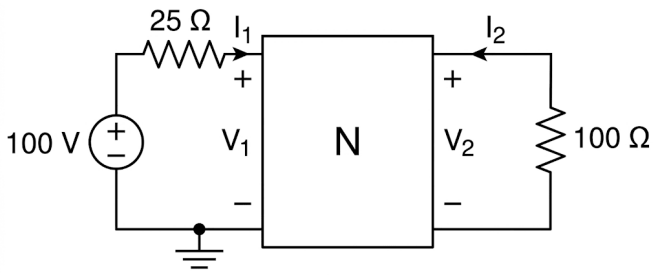


Fig. Q6.11.1

A. 1/90 **B.** -1/90 **C.** -1/99 **D.** -1/11

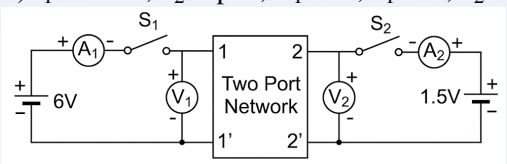
Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions 6.08.1 and 6.08.2 A two-port network shown below is excited by external dc sources. The voltages and the currents are measured with voltmeters V_1,V_2 and ammeters A_1, A_2 (all assumed to be ideal), as indicated. Under following switch conditions, the readings obtained are :

i)S_1-Open, S_2-Closed; A_1=0A, V_1=4.5V, V_2=1.5V, A_2=1A

ii)S_1-Closed, S_2-Open; A_1=4A, V_1=6V, V_2=6V, A_2=0A



Q6.08.1 MCQ 2008 2M

The z-parameter matrix for this network is

A. $\begin{bmatrix}1.5 & 1.5\\ 4.5 & 1.5\end{bmatrix}$ **B.** $\begin{bmatrix}1.5 & 4.5\\ 1.5 & 4.5\end{bmatrix}$

C. $\begin{bmatrix}1.5 & 4.5\\ 1.5 & 1.5\end{bmatrix}$ **D.** $\begin{bmatrix}4.5 & 1.5\\ 1.5 & 4.5\end{bmatrix}$

Q6.08.2 MCQ 2008 2M

The h-parameter matrix for this network is

A. $\begin{bmatrix}-3 & 3\\ -1 & 0.67\end{bmatrix}$ **B.** $\begin{bmatrix}-3 & -1\\ 3 & 0.67\end{bmatrix}$

C. $\begin{bmatrix}-3 & 3\\ 1 & 0.67\end{bmatrix}$ **D.** $\begin{bmatrix}3 & 1\\ -3 & 0.67\end{bmatrix}$

Q6.06.1 MCQ 2006 2M

In the two-port network shown in the figure below, z_{12} and z_{21} are, respectively

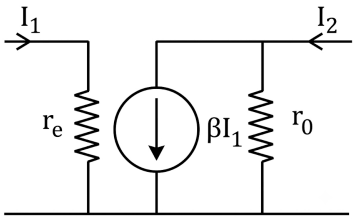


Fig. Q6.06.1

A. r_c and βr_0 **B.** 0 and $-\beta r_0$

C. 0 and βr_0 **D.** r_c and $-\beta r_0$

Q6.06.2 MCQ 2006 2M

A two port network is represented by ABCD parameters given by $\begin{bmatrix}V_1\\ I_1\end{bmatrix} = \begin{bmatrix}A & B\\ C & D\end{bmatrix}\begin{bmatrix}V_2\\ -I_2\end{bmatrix}$. If port-2 is terminated by R_L, the input impedance seen at port-1 is given by

A. $\dfrac{A+BR_L}{C+DR_L}$ **B.** $\dfrac{AR_L+C}{BR_L+D}$

C. $\dfrac{DR_L+A}{BR_L+C}$ **D.** $\dfrac{B+AR_L}{D+CR_L}$

Q6.05.1 MCQ 2005 2M

The h-parameters of the circuit shown in figure are

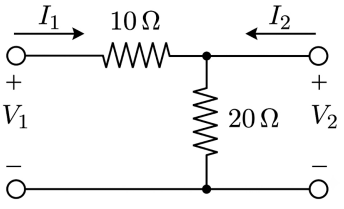


Fig. Q6.05.1

A. $\begin{bmatrix}0.1 & 0.1\\ -0.1 & 0.3\end{bmatrix}$ **B.** $\begin{bmatrix}10 & -1\\ 1 & 0.05\end{bmatrix}$

C. $\begin{bmatrix}30 & 20\\ 20 & 20\end{bmatrix}$ **D.** $\begin{bmatrix}10 & 1\\ -1 & 0.05\end{bmatrix}$

Q6.05.2 MCQ 2005 1M

The ABCD parameters of an ideal n:1 transformer shown in figure are $\begin{bmatrix}n & 0\\ 0 & X\end{bmatrix}$. The value of X will be

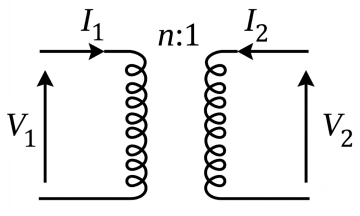


Fig. Q6.05.2

A. n **B.** $\frac{1}{n}$ **C.** n^2 **D.** $\frac{1}{n^2}$

Q6.04.1 MCQ 2004 2M

For the lattice circuit shown in Figure, $Z_a = j2\Omega$ and $Z_b = 2\Omega$. The values of the open circuit impedance parameters $[z] = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$ are

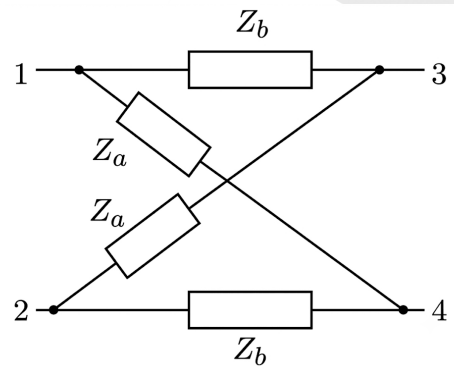


Fig. Q6.04.1

A. $\begin{bmatrix} 1-j & 1+j \\ 1+j & 1+j \end{bmatrix}$ **B.** $\begin{bmatrix} 1-j & 1+j \\ 1+j & 1-j \end{bmatrix}$

C. $\begin{bmatrix} 1+j & 1+j \\ 1-j & 1-j \end{bmatrix}$ **D.** $\begin{bmatrix} 1+j & -1+j \\ -1+j & 1+j \end{bmatrix}$

Q6.03.1 MCQ 2003 2M

The impedance parameters z_{11} and z_{12} of the two-port network in figure are

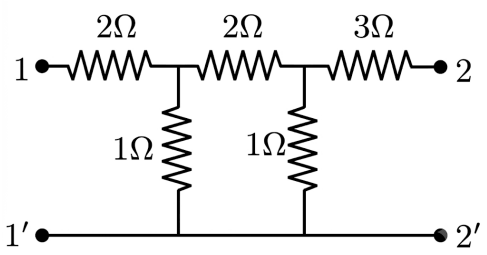


Fig. Q6.03.1

A. $z_{11} = 2.75\Omega$ and $z_{12} = 0.25\Omega$

B. $z_{11} = 3\Omega$ and $z_{12} = 0.5\Omega$

C. $z_{11} = 3\Omega$ and $z_{12} = 0.25\Omega$

D. $z_{11} = 2.25\Omega$ and $z_{12} = 0.5\Omega$

Q6.02.1 NAT 2002 5M

Consider the network in Figure

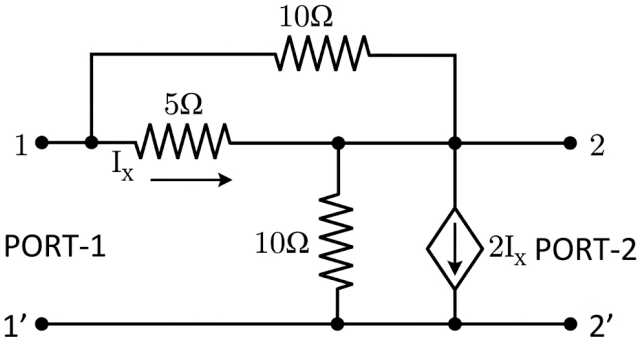


Fig. Q6.02.1

(a) Find its short-circuit admittance parameters.
(b) Find the open-circuit impedance z_{22}.

Q6.01.1 MCQ 2001 1M

The admittance parameter Y_{12} in the 2-port network in Figure is

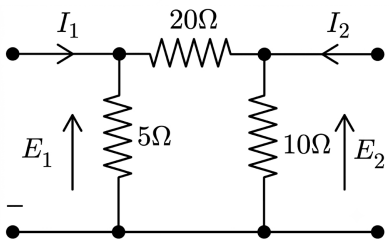


Fig. Q6.01.1

A. -0.2 mho **B.** 0.1 mho
C. -0.05 mho **D.** 0.05 mho

Q6.01.2 MCQ 2001 2M

The z parameters Z_{11} and Z_{21} for the 2-port network in figure are

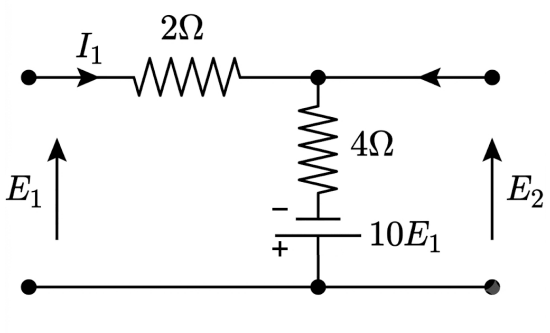


Fig. Q6.01.2

A. $z_{11} = -\frac{6}{11}\Omega; z_{21} = \frac{16}{11}\Omega$

B. $z_{11} = \frac{6}{11}\Omega; z_{21} = \frac{4}{11}\Omega$

C. $z_{11} = \frac{6}{11}\Omega; z_{21} = -\frac{16}{11}\Omega$

D. $z_{11} = \frac{4}{11}\Omega; z_{21} = \frac{4}{11}\Omega$

Q6.01.3 NAT 2001 5M

The admittance parameters of a 2-port network shown in figure are given by $Y_{11} = 2$ mho, $Y_{12} = -0.5$ mho, $Y_{21} = 4.8$ mho, $Y_{22} = 1$ mho. The output port is terminated with a load admittance $Y_L = 0.2$ mho. Find E_2 for each of the following conditions?

(a) $E_1 = 10\angle 0°$ V

(b) $I_1 = 10\angle 0°$ A

(c) A source $10\angle 0°$ V in series with a 0.25Ω resistor is connected to the input port.

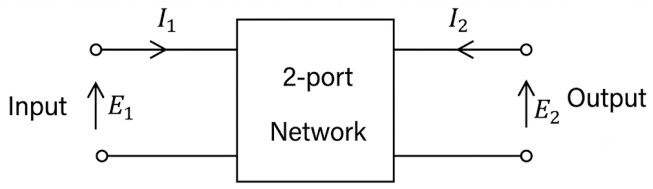


Fig. Q6.01.3

Q6.99.1 MCQ 1999 1M

A 2-port network is shown in Figure. The parameters h_{21} for this network can be given by

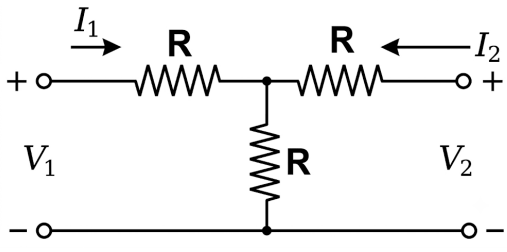


Fig. Q6.99.1

A. $-\frac{1}{2}$ **B.** $+\frac{1}{2}$ **C.** $-\frac{3}{2}$ **D.** $+\frac{3}{2}$

Q6.98.1 MCQ 1998 1M

The short-circuit admittance matrix of a two-port network is $\begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$. The two port network is

A. Non-reciprocal and passive

B. Non-reciprocal and active

C. Reciprocal and passive

D. Reciprocal and active

Q6.94.1 NAT 1994 1M

The condition that a 2-port network is reciprocal, can be expressed in terms of its ABCD parameters as _____.

Q6.93.1 NAT 1993 2M

Find the Y-parameters (short circuit admittance parameters) for the network shown in figure.

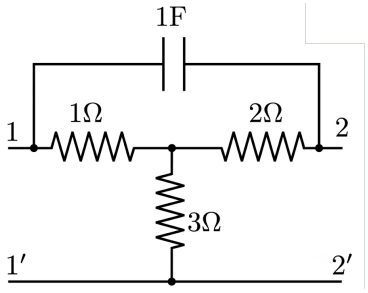


Fig. Q6.93.1

Q6.92.1 MSQ 1992 2M

For a two port network to be reciprocal

☐ **A.** $z_{11} = z_{22}$

☐ **B.** $y_{21} = y_{12}$

☐ **C.** $h_{21} = -h_{12}$

☐ **D.** AD-BC=0

Q6.91.1 MCQ 1991 1M

Two two port networks are connected in cascade. The combination is to be represented as a single two-port network. The parameters of the network are obtained by multiplying the individual

A. z-parameter matrix

B. h-parameter matrix

C. y-parameter matrix

D. ABCD parameter matrix

Q6.90.1 MCQ 1990 2M

The open circuit impedance matrix of the 2 port network shown in figure is

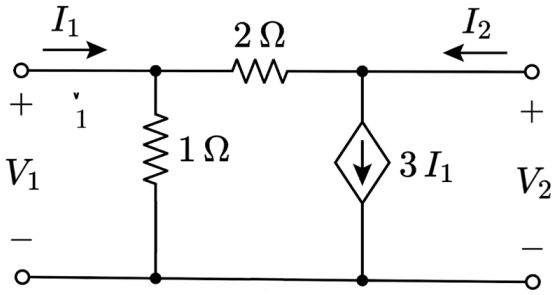


Fig. Q6.90.1

A. $\begin{bmatrix} -2 & 1 \\ -8 & 3 \end{bmatrix}$ **B.** $\begin{bmatrix} -2 & -8 \\ -8 & 3 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **D.** $\begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

Q6.89.1 MSQ 1989 1M

For the transfer function of a physical two-port network

☐ **A.** all the zeroes must lie only in the left half of the s-plane

☐ **B.** the poles may lie anywhere in the s-plane

☐ **C.** the poles lying on the imaginary axis must be simple

☐ **D.** a pole may lie at origin

Q6.89.2 MCQ 1989 1M

The condition AD-BC=1 for a two-port network implies that the network is a

A. reciprocal network

B. lumped element network

C. lossless network

D. unilateral element network

Q6.88.1 MCQ 1988 1M

Two two port networks are connected in parallel. The combination is to be represented as a single two-port network. The parameters of the network are obtained by addition of the individual

A. z-parameters

B. h-parameters

C. y-parameters

D. ABCD parameters

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q6.26.1 D	Q6.25.1 A	Q6.24.1 1.5 to 1.5
Q6.23.1 80 to 80	Q6.23.2 MTA	Q6.22.1 3.9 to 4.1
Q6.21.1 C	Q6.20.1 48 to 48	Q6.18.1 4.3 to 5.3
Q6.16.1 D	Q6.16.2 2.98 to 3.02	Q6.16.3 A
Q6.15.1 *	Q6.15.2 B	Q6.14.1 1.24 to 1.26
Q6.14.2 C	Q6.12.1 C	Q6.12.2 C

Q6.11.1 D	Q6.08.1 C	Q6.08.2 A
Q6.06.1 B	Q6.06.2 D	Q6.05.1 D
Q6.05.2 B	Q6.04.1 D	Q6.03.1 A
Q6.02.1 *	Q6.01.1 C	Q6.01.2 C
Q6.01.3 *	Q6.99.1 A	Q6.98.1 B
Q6.94.1 *	Q6.93.1 *	Q6.92.1 B,C
Q6.91.1 D	Q6.90.1 A	Q6.89.1 C,D
Q6.89.2 A	Q6.88.1 C	

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